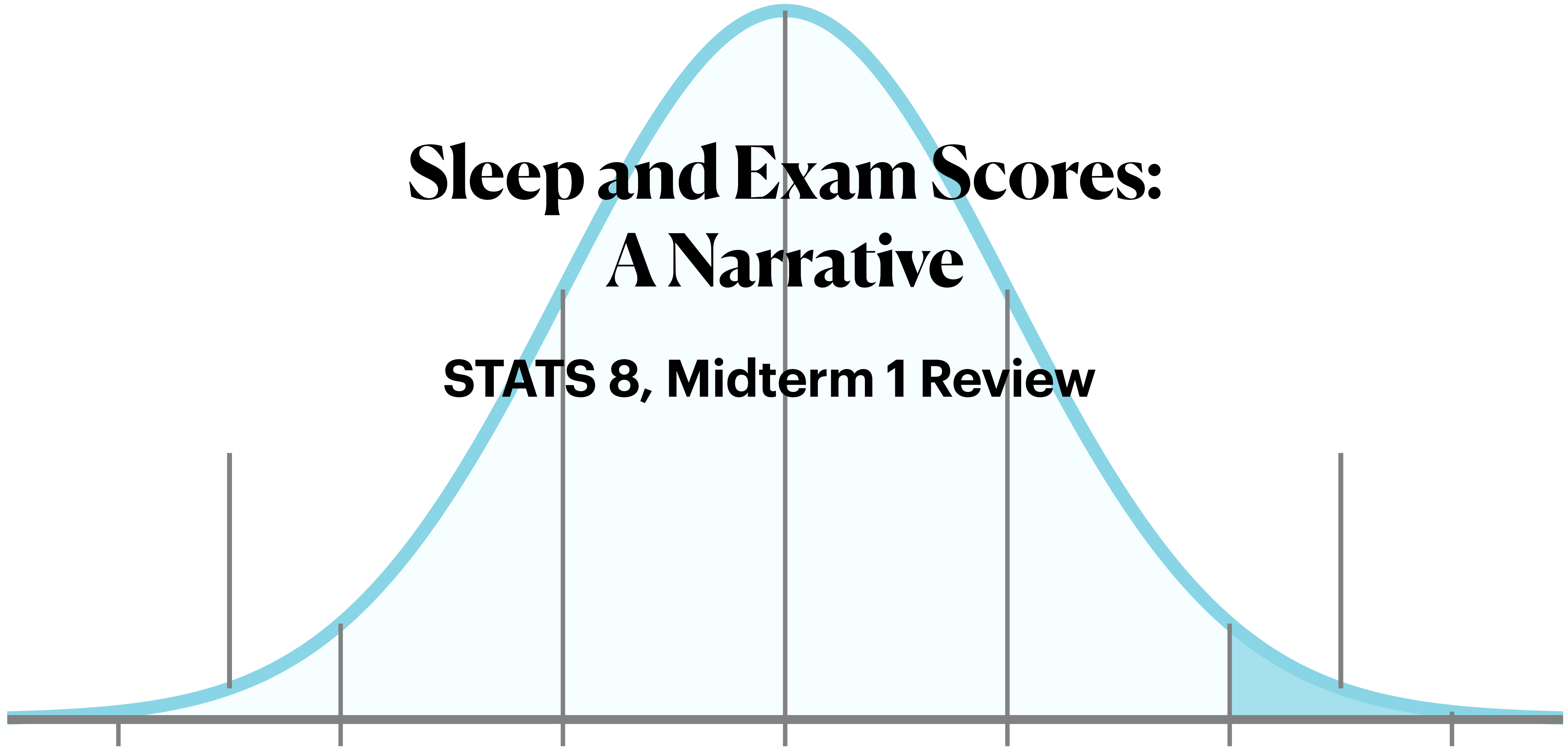


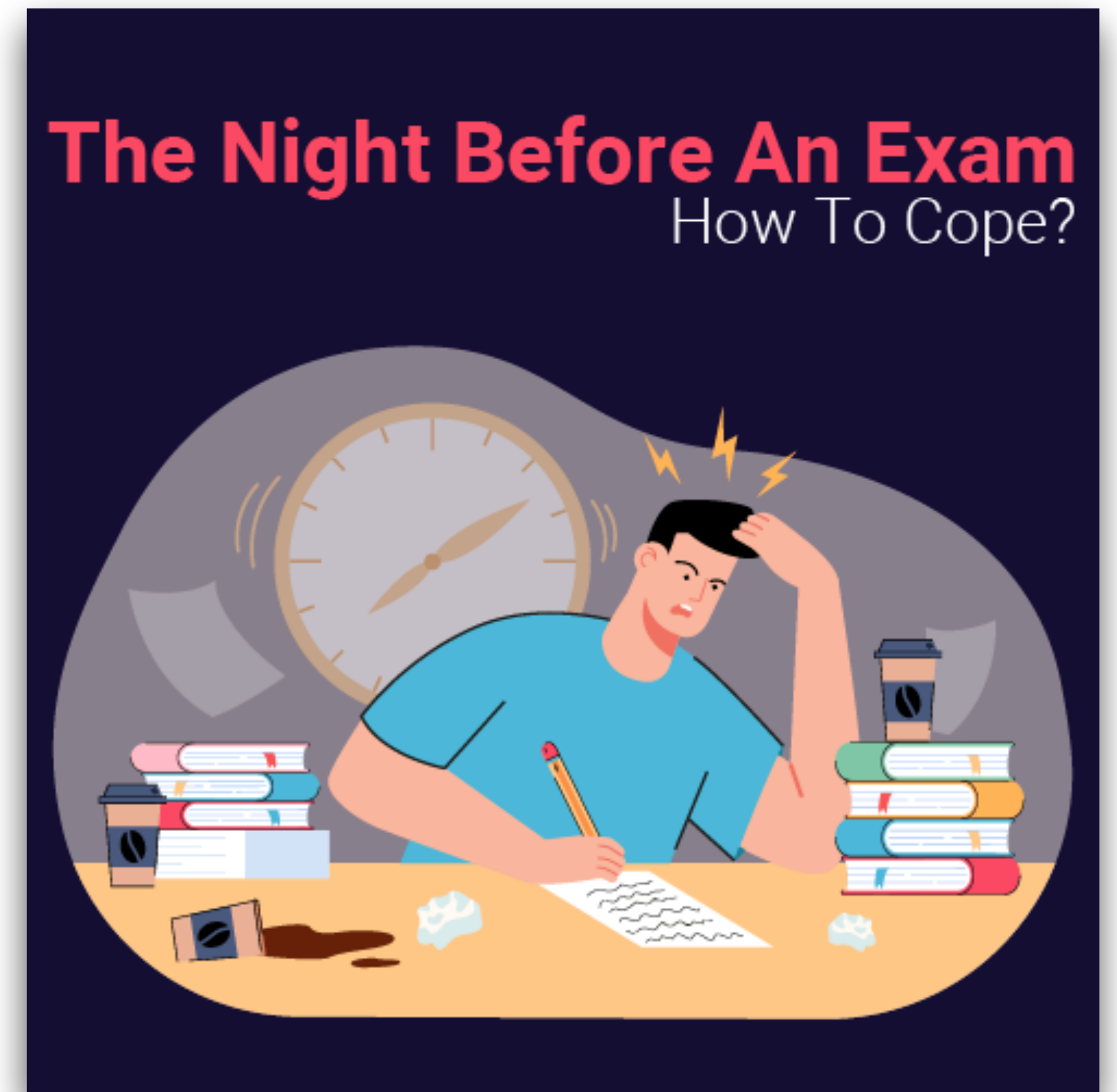
Sleep and Exam Scores: A Narrative

STATS 8, Midterm 1 Review



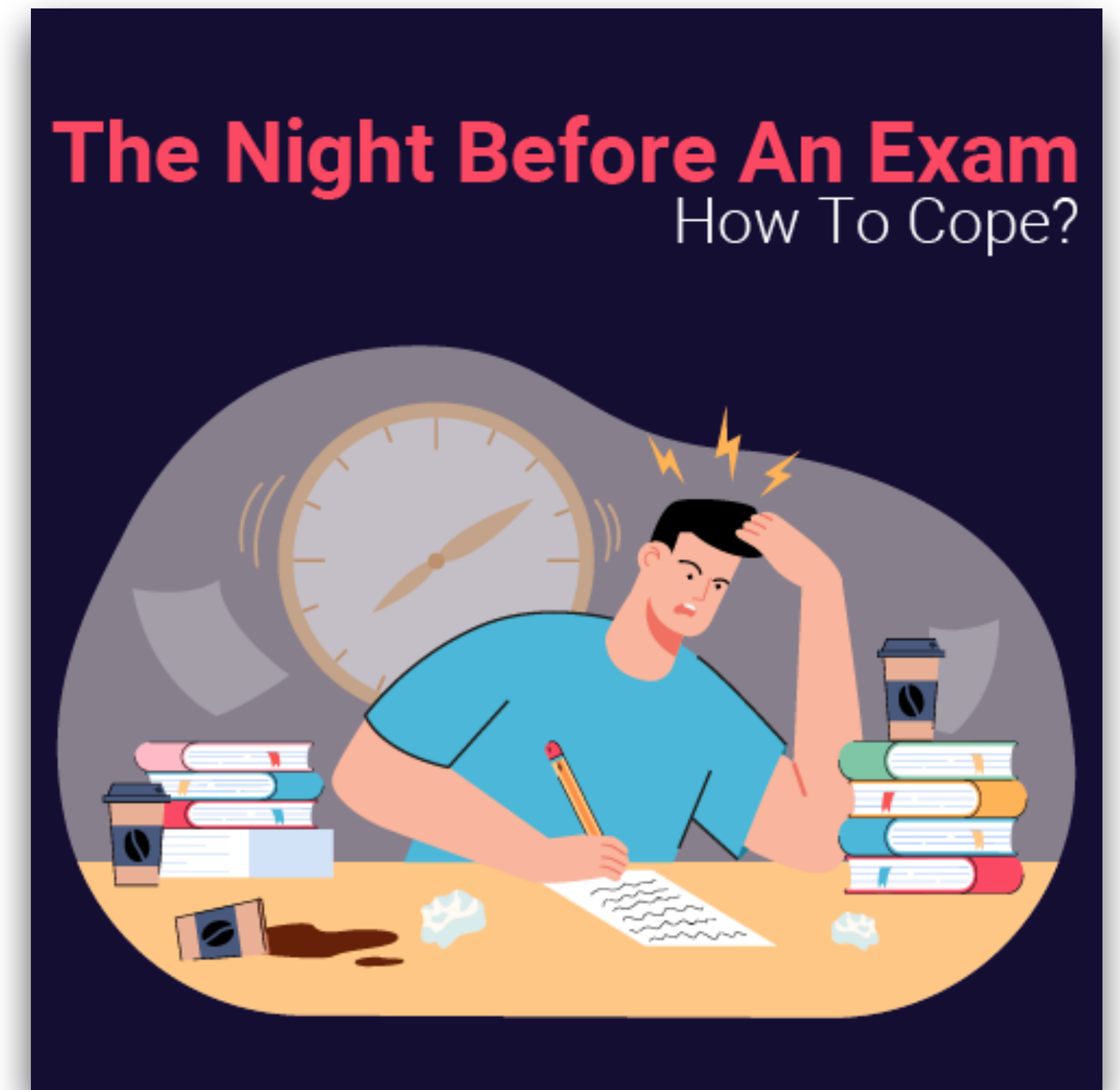
Scientific Question of Interest

- My friend Winter is teaching STATS 7, and we are both giving exams soon. We start wondering whether students should be told to prioritize sleep before an exam. So we begin with the question:
 - Does getting more sleep increase undergraduate students' exam scores?
- This is where the study begins: not with data, but with a **scientific question** about a **population of interest**.



Scientific Question of Interest

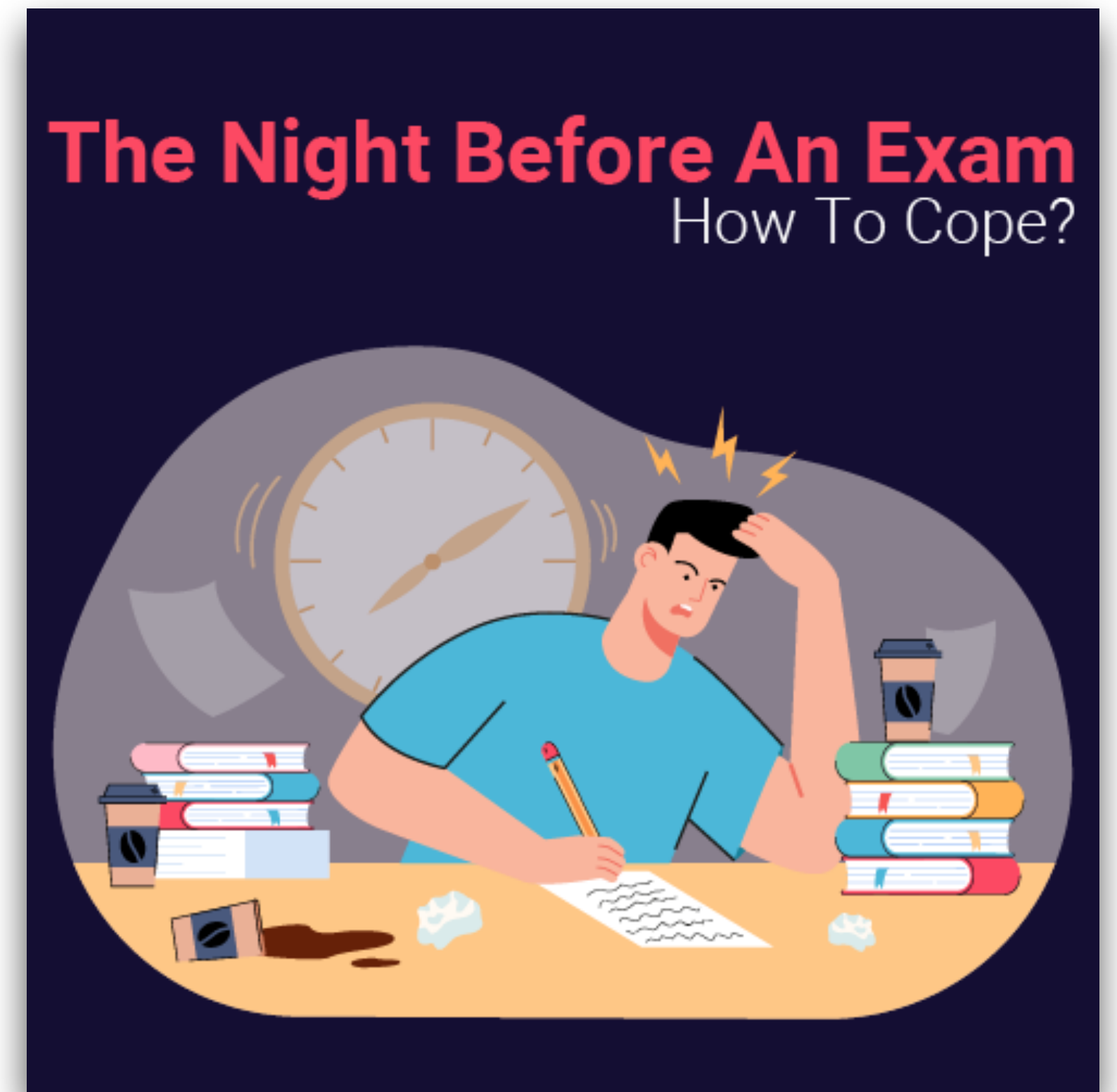
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 1. Who is our population of interest?



Scientific Question of Interest

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- This is where the study begins: not with data, but with a **scientific question** about a **population of interest**.
 1. Who is our population of interest?

undergraduate students



Data Collection & Study Design

- Does getting more sleep increase undergraduate students' exam scores?
- Winter points out that we asked, "Does getting more sleep *increase* exam scores?" That is a causal question. We are not just asking whether sleep and score are related, we are asking whether changing sleep would change score.
- 2. How does this affect the kind of study we need?



Data Collection & Study Design

- Does getting more sleep increase undergraduate students' exam scores?
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- 2. How does this affect the kind of study we need?

Needs to be a randomized experiment!



Data Collection & Study Design

- 📌 Does getting more sleep increase undergraduate students' exam scores?
 - If we truly want to study causation, Winter says we would need an **experiment**.
3. What does that mean?

Data Collection & Study Design

- 📌 Does getting more sleep increase undergraduate students' exam scores?
 - If we truly want to study causation, Winter says we would need an **experiment**.
3. What does that mean?

Students need to be randomly assigned to get different amounts of sleep before their exam.

Data Collection & Study Design

-  Does getting more sleep increase undergraduate students' exam scores?
- If we truly want to study causation, Winter says we would need an **experiment**. In an experiment, researchers impose treatments on the observational units.
 - So here, that would mean assigning students to sleep different amounts before an exam and then recording their scores.
- 4. What do you think about this idea?

Data Collection & Study Design

- 📌 Does getting more sleep increase undergraduate students' exam scores?
- If we truly want to study causation, Winter says we would need an **experiment**. In an experiment, researchers impose treatments on the observational units.
 - So here, that would mean assigning students to sleep different amounts before an exam and then recording their scores.
- But once we say that out loud, it sounds ridiculous. **We cannot realistically force students into sleep schedules right before a midterm, and even if we tried, compliance would be messy and ethics would be questionable.** So although an experiment would be the right design for causation in principle, it is not a good plan here.

Data Collection & Study Design

~~📌 Does getting more sleep increase undergraduate students' exam scores?~~

★ Is getting more sleep associated with higher exam scores in undergraduate students?

- Winter and I are satisfied that this is a question we can answer. Now, we need to get a sample of students to learn about. I suggest walking around in the library late one evening and asking students about their last midterm.
5. Sampling students in the library at night is best described as...



Data Collection & Study Design

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5. Sampling students in the library at night is best described as...

convenience sampling



Data Collection & Study Design

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6. This would likely cause...



Data Collection & Study Design

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- ★ Is getting more sleep associated with higher exam scores in undergraduate students?
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- 6. This would likely cause...

Selection bias



Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

- Winter says that this is a **convenience sample**, and we'll definitely have **selection bias**. She suggests instead using probability sampling.

7. What is probability sampling?

Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

- Winter says that this is a **convenience sample**, and we'll definitely have **selection bias**. She suggests instead using probability sampling.

7. What is probability sampling?

**A sampling method in which individuals have pre-specified/
known probabilities of being selected**

Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

- I like the idea of probability sampling, but I worry about logistics. We probably do not have a neat list of every undergraduate student to sample from. So Winter suggests a practical alternative: randomly select undergraduate courses from the course schedule, contact the instructors, and ask them to give a short survey to all students in those classes after the exam.
8. What type of sampling best describes this approach?

Data Collection & Study Design

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Cluster sampling

Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses

- At this point, we should be precise in recording our procedure.

9. What is the **sampling frame**?

the list of current UCI undergraduate courses in the course schedule

Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

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9. What is the **sampling frame**?

Data Collection & Study Design

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- At this point, we should be precise in recording our procedure.

10. What are the **observational units**?

Data Collection & Study Design

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- At this point, we should be precise in recording our procedure.

10. What are the **observational units**?

the individual students surveyed

Data Collection & Study Design

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses

- Now, we need to decide what goes on the survey. We know we need to ask about sleep and exam score, and it's probably also a good idea to record which class the student's exam was in. Winter writes up a survey that looks like this:

Name:

ID:

Score on midterm in this class:

Amount of sleep (hours) the night before the midterm:

11. What type of bias is most likely if students are asked to report their own exam scores on a non-anonymous survey?

Data Collection & Study Design

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Name:

ID:

Score on midterm in this class:

Amount of sleep (hours) the night before the midterm:

Response bias

11. What type of bias is most likely if students are asked to report their own exam scores on a non-anonymous survey?

Data Description

- ★ Is getting more sleep associated with higher exam scores in undergraduate students?
- 📌 Data collection: cluster sampling of undergraduate courses, anonymous survey
- I change Winter's survey to be anonymous, since we don't need students' IDs anyway. We administer the survey and receive our results. The raw data looks like this:

class	sleep_hours	exam_score
Physics 7C	7.1	98
Math 2B	5.8	75
Bio Sci 93	5	58
Math 2B	5.8	83
Philosophy 3	6.8	81
Sociology 1	5.1	44
Sociology 1	7.2	86
Anthro 2A	7.3	80

12. What types of variables are these?

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Sociology 1	7.2	86
Anthro 2A	7.3	80

12. What types of variables are these?

**Categorical (nominal),
quantitative, quantitative**

Data Description

- ★ Is getting more sleep associated with higher exam scores in undergraduate students?
 - 📌 Data collection: cluster sampling of undergraduate courses, anonymous survey
 - We finally have the survey results. Before jumping into relationships, Winter reminds me that we should first understand each variable on its own. Sleep hours and exam score are both quantitative, so we start with numerical summaries of center and spread.
13. Which of the following would be reasonable numerical summaries for sleep hours?

Data Description

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13. Which of the following would be reasonable numerical summaries for sleep hours?

Mean, median, IQR, variance, range

Data Description

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14. Which variable would it *not* make sense to summarize using a mean?

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14. Which variable would it *not* make sense to summarize using a mean?

Class

Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- Winter and I compute some summary statistics using R.

- Exam score:

- Mean: 76.06, IQR: 16, Range: 58

- Sleep hours:

- Mean: 6.64, IQR: 1.8, Range: 418

15. Does anything stand out to you here?

Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- I'm concerned about sleep hours having a **range of 418**. Something's not right there. I decide to plot the observations to understand what's happening visually.

16. Which of the following graphical displays would be appropriate to summarize sleep hours?

Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

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- I'm concerned about sleep hours having a **range of 418**. Something's not right there. I decide to plot the observations to understand what's happening visually.

16. Which of the following graphical displays would be appropriate to summarize sleep hours?

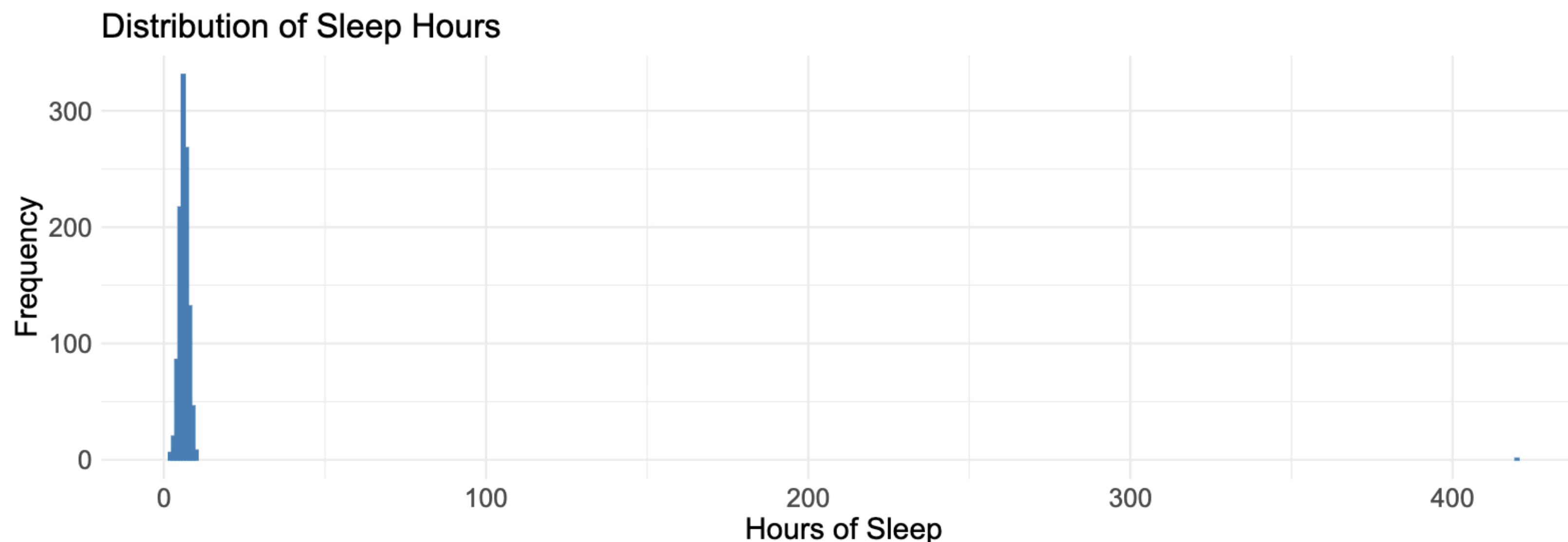
Boxplot, histogram

Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- I'm concerned about sleep hours having a range of 418. Something's not right there. I decide to plot the observations to understand what's happening visually.
- I start with a histogram, pictured below.

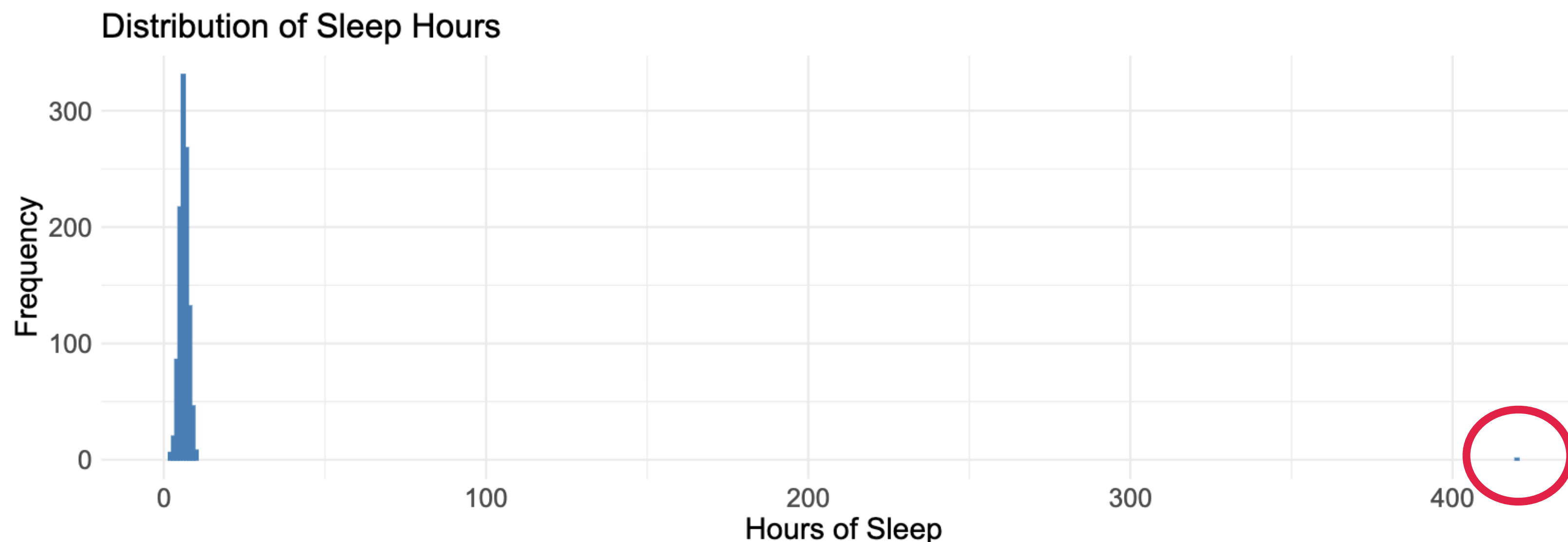


Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

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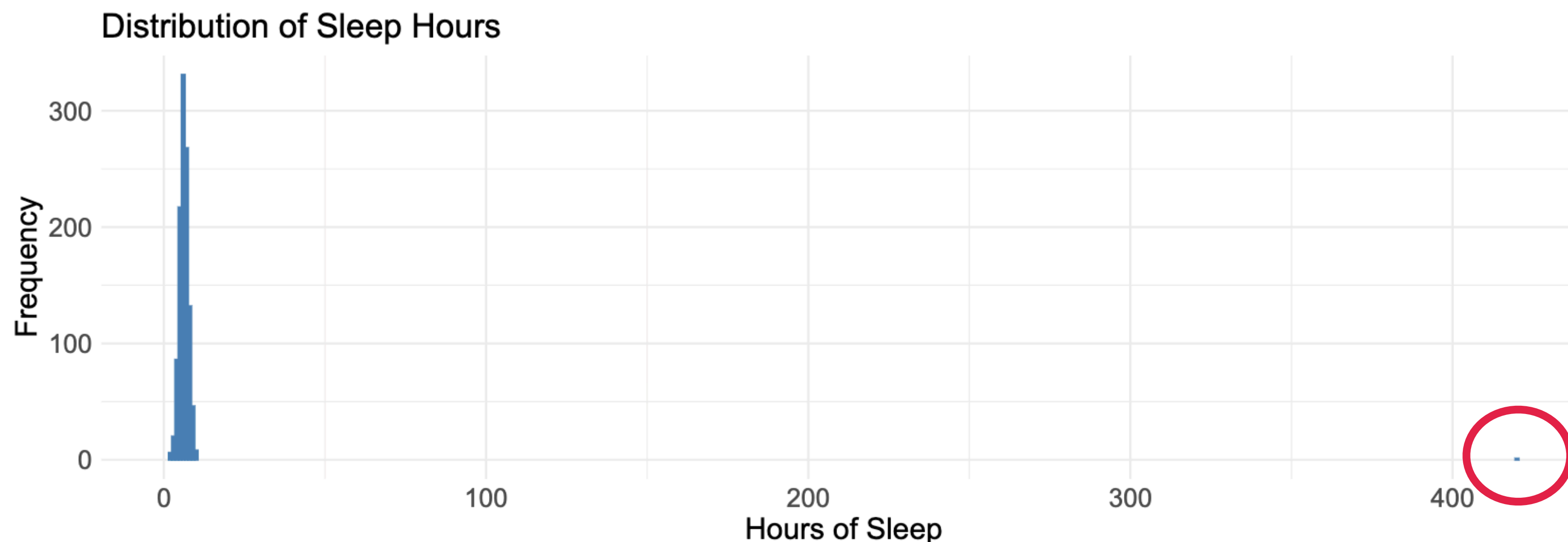
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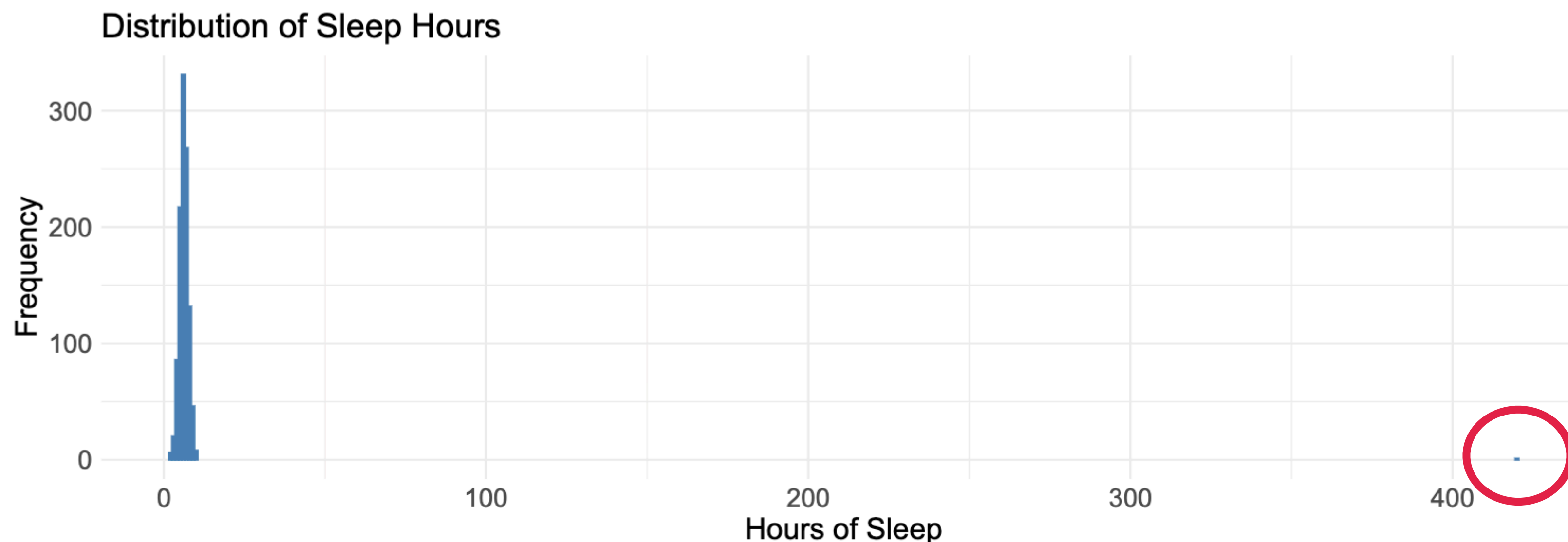
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17. What would you do with this outlier?



Data Description

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 - 📌 Data collection: cluster sampling of undergraduate courses, anonymous survey
 - I'm concerned about sleep hours having a range of 418. Something's not right there. I decide to plot the observations to understand what's happening visually.
17. What would you do with this outlier? **It is impossible, a mistake — discard.**

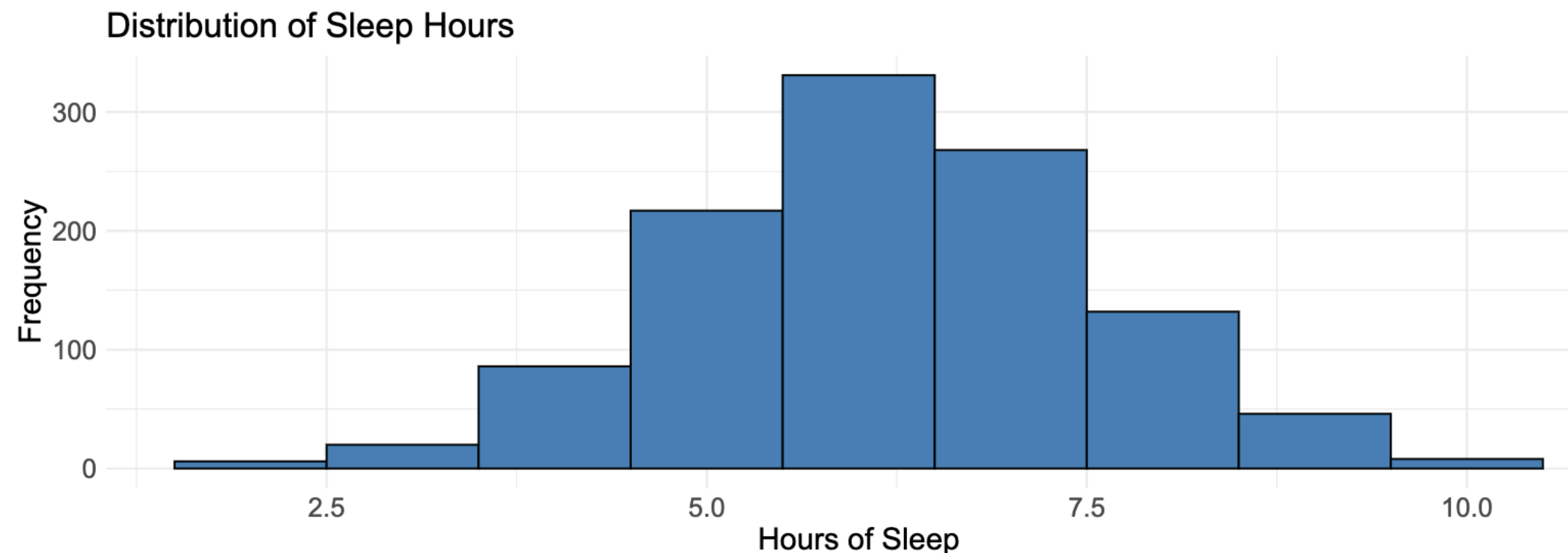


Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

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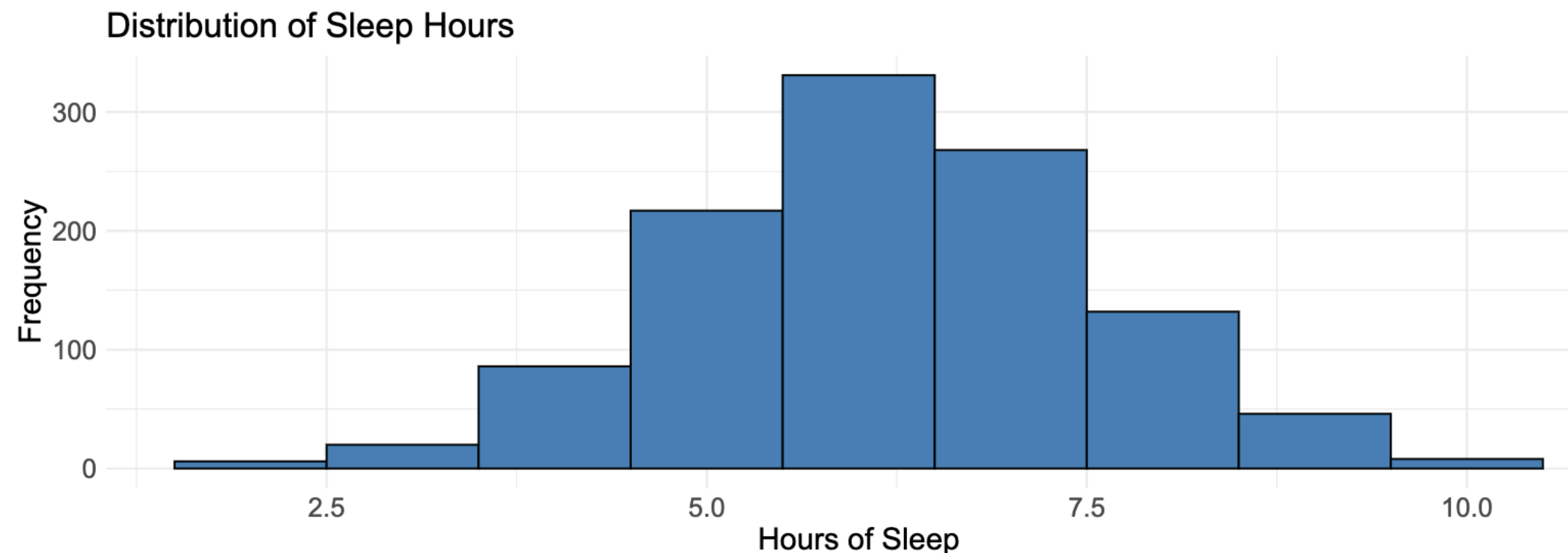
- I'm concerned about sleep hours having a range of 418. Something's not right there. I decide to plot the observations to understand what's happening visually.
- Mistake removed!



Data Description

- ★ Is getting more sleep associated with higher exam scores in undergraduate students?
- 📌 Data collection: cluster sampling of undergraduate courses, anonymous survey
- I'm concerned about sleep hours having a range of 418. Something's not right there. I decide to plot the observations to understand what's happening visually.

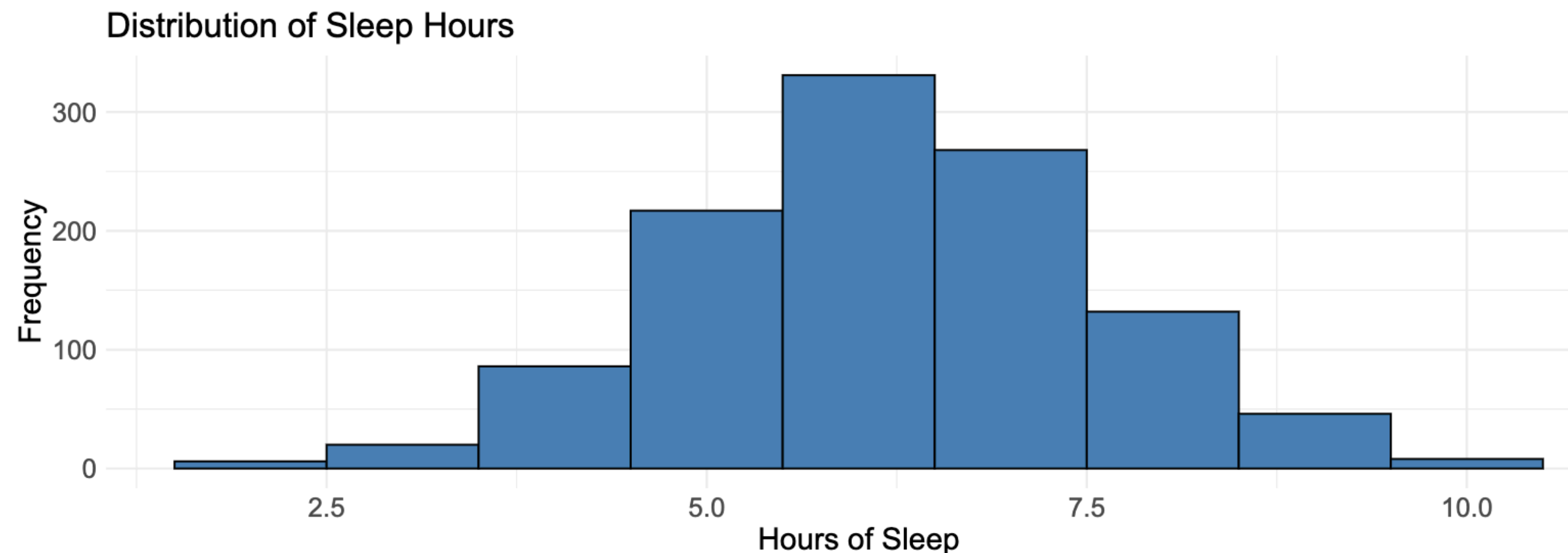
18. How might you describe the shape of this distribution?



Data Description

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18. How might you describe the shape of this distribution? **(Roughly) symmetric**

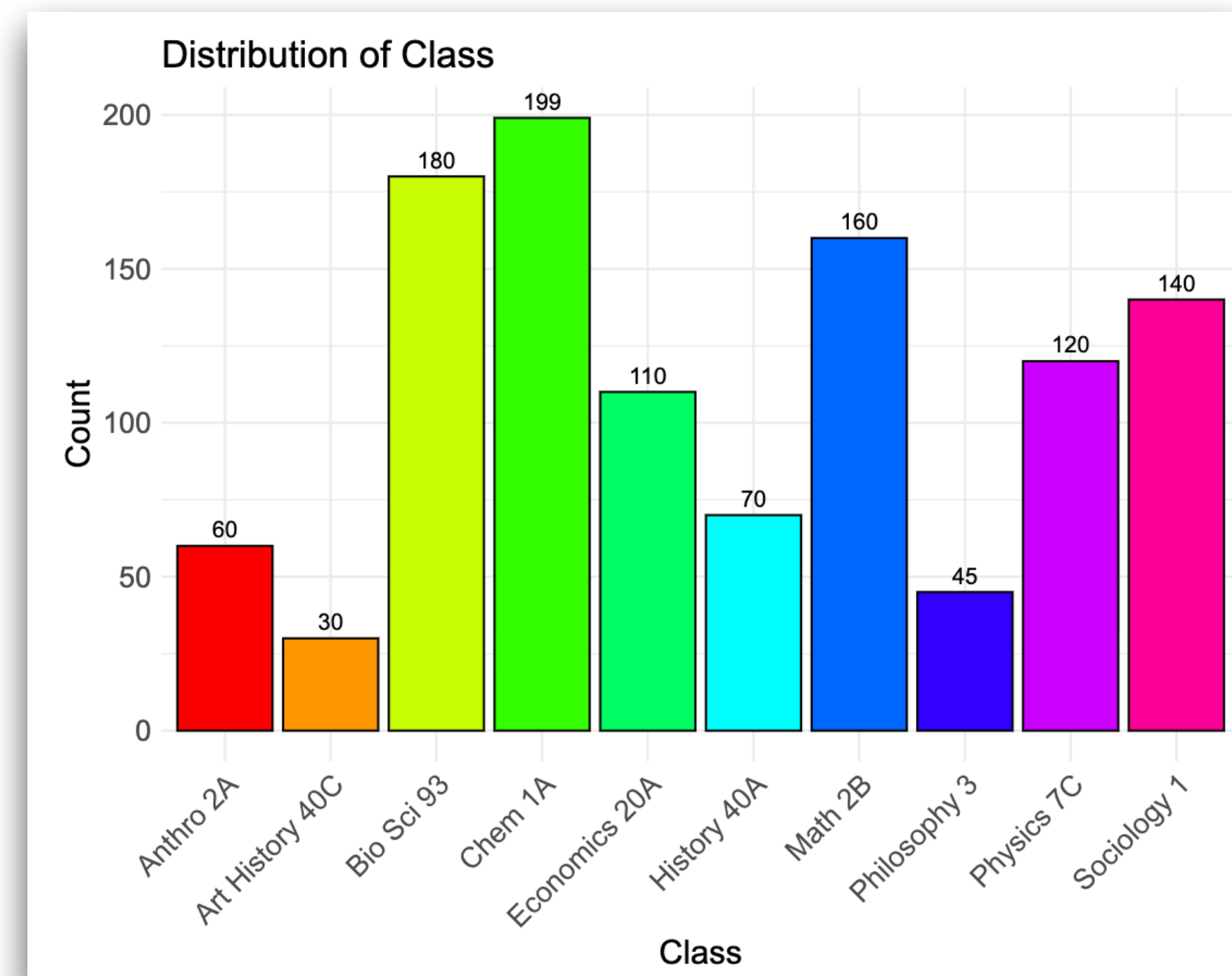
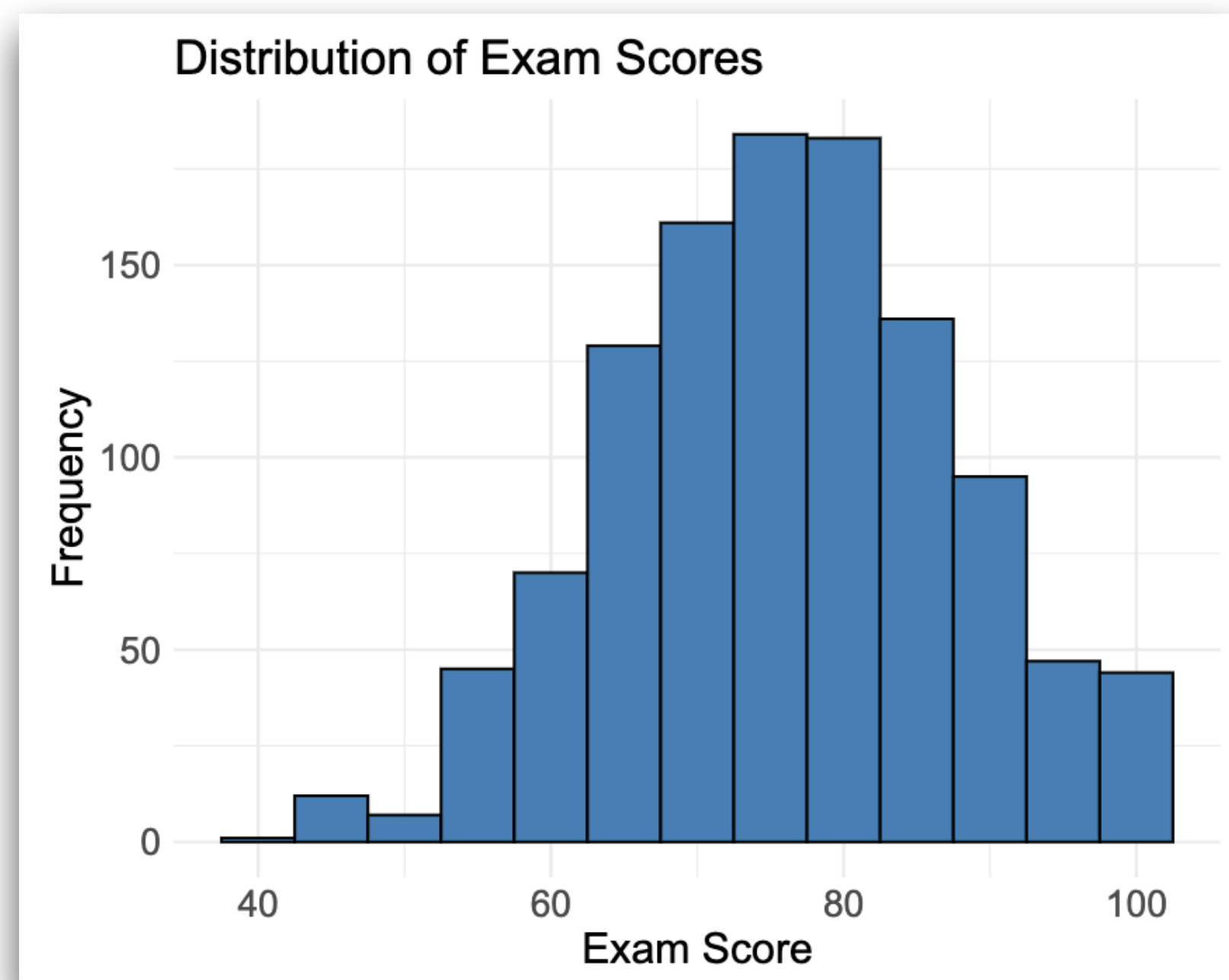


Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- Now that we have identified the mistake in sleep hours, we want to do some summaries of exam score and class:

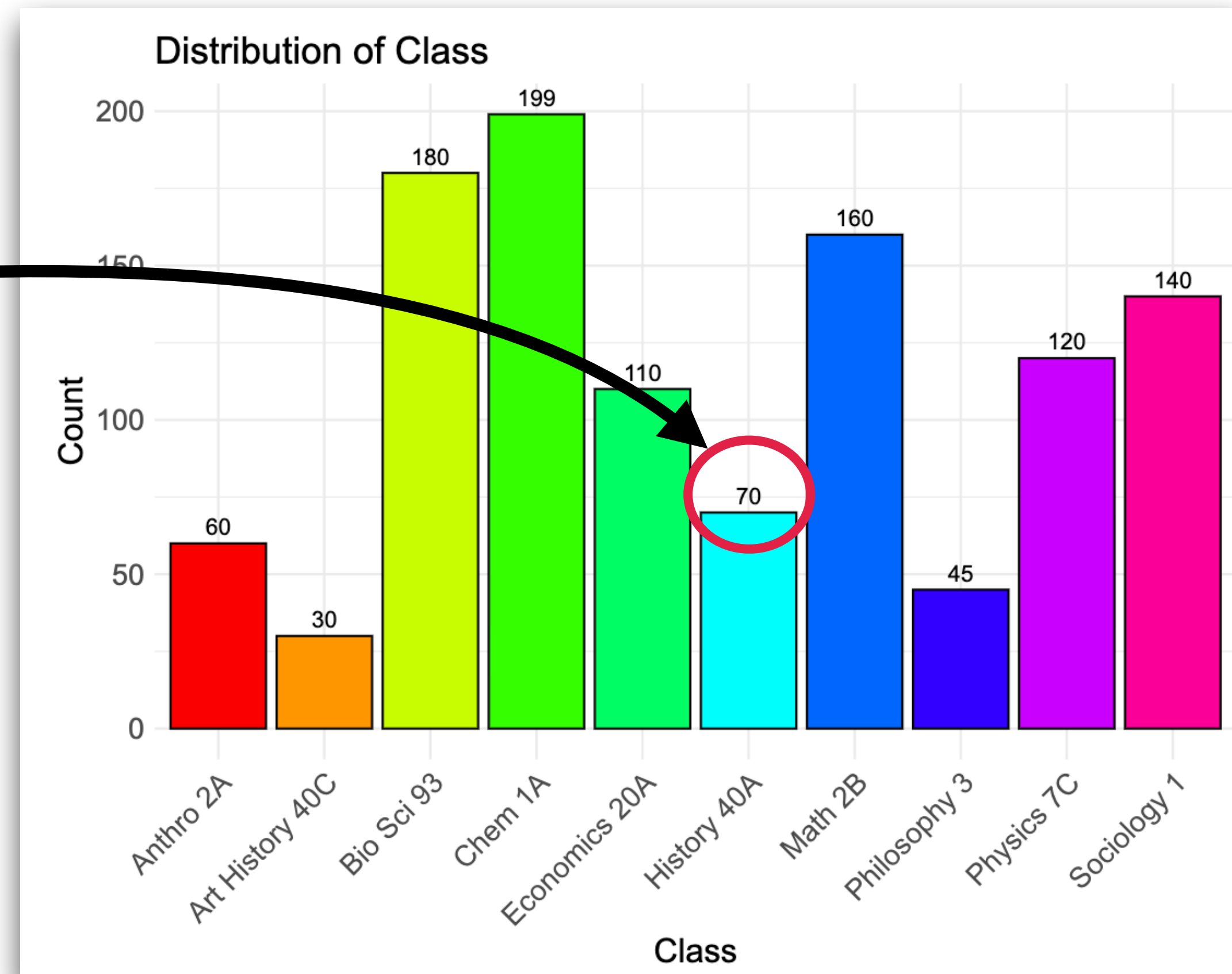


Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

19. What is the interpretation of this value, 70?



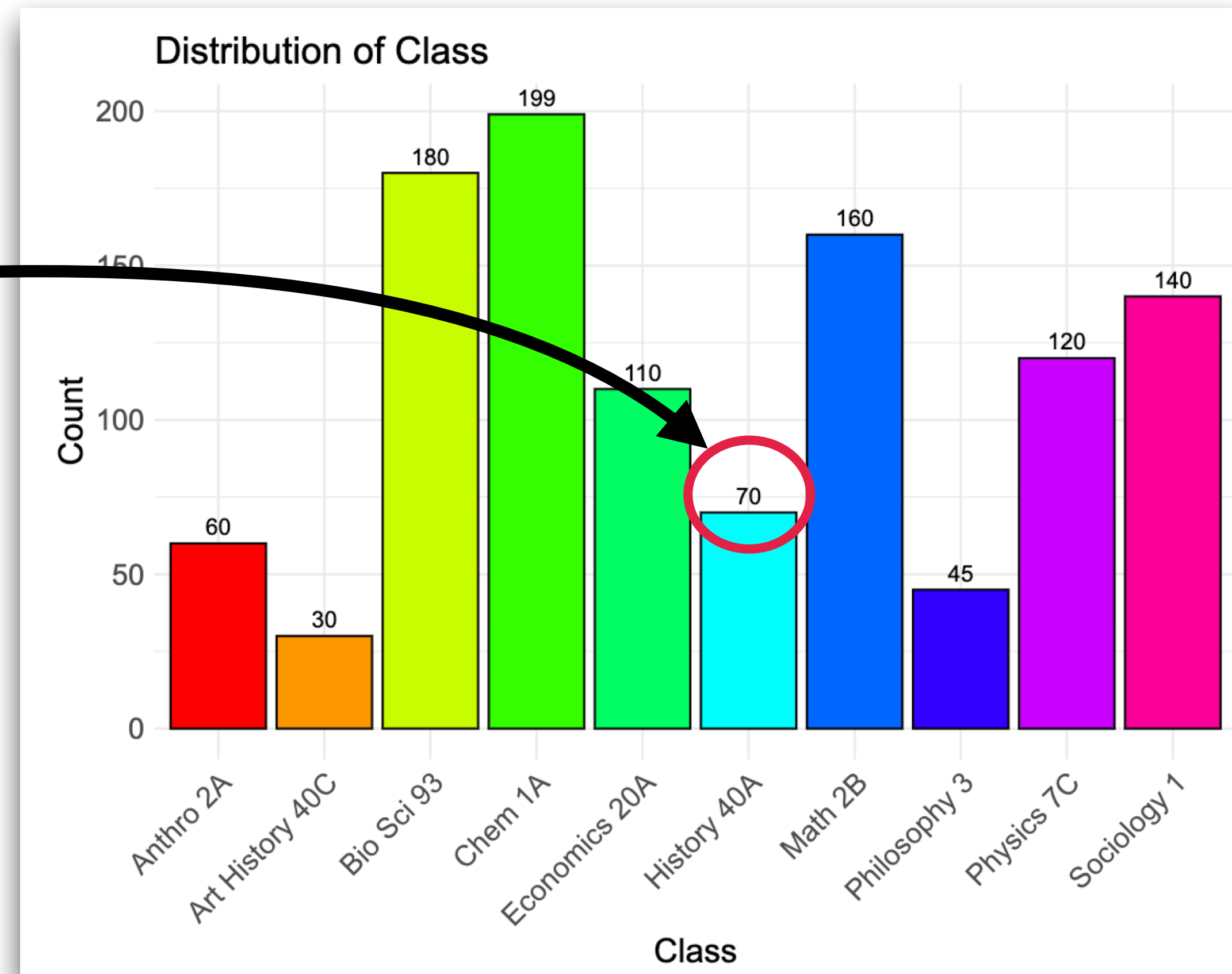
Data Description

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

19. What is the interpretation of this value, 70?

70 students who filled out this survey were in History 40A.

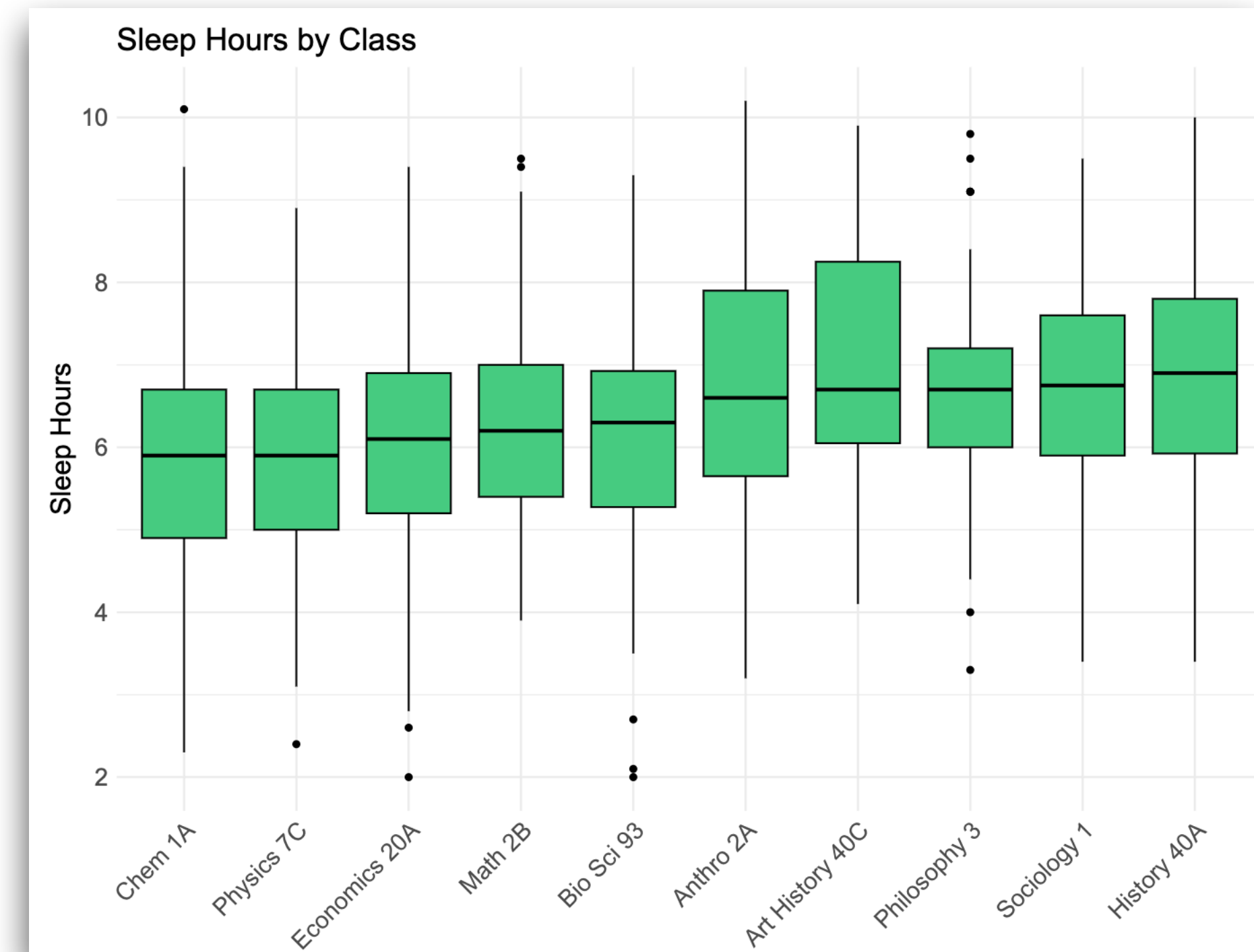
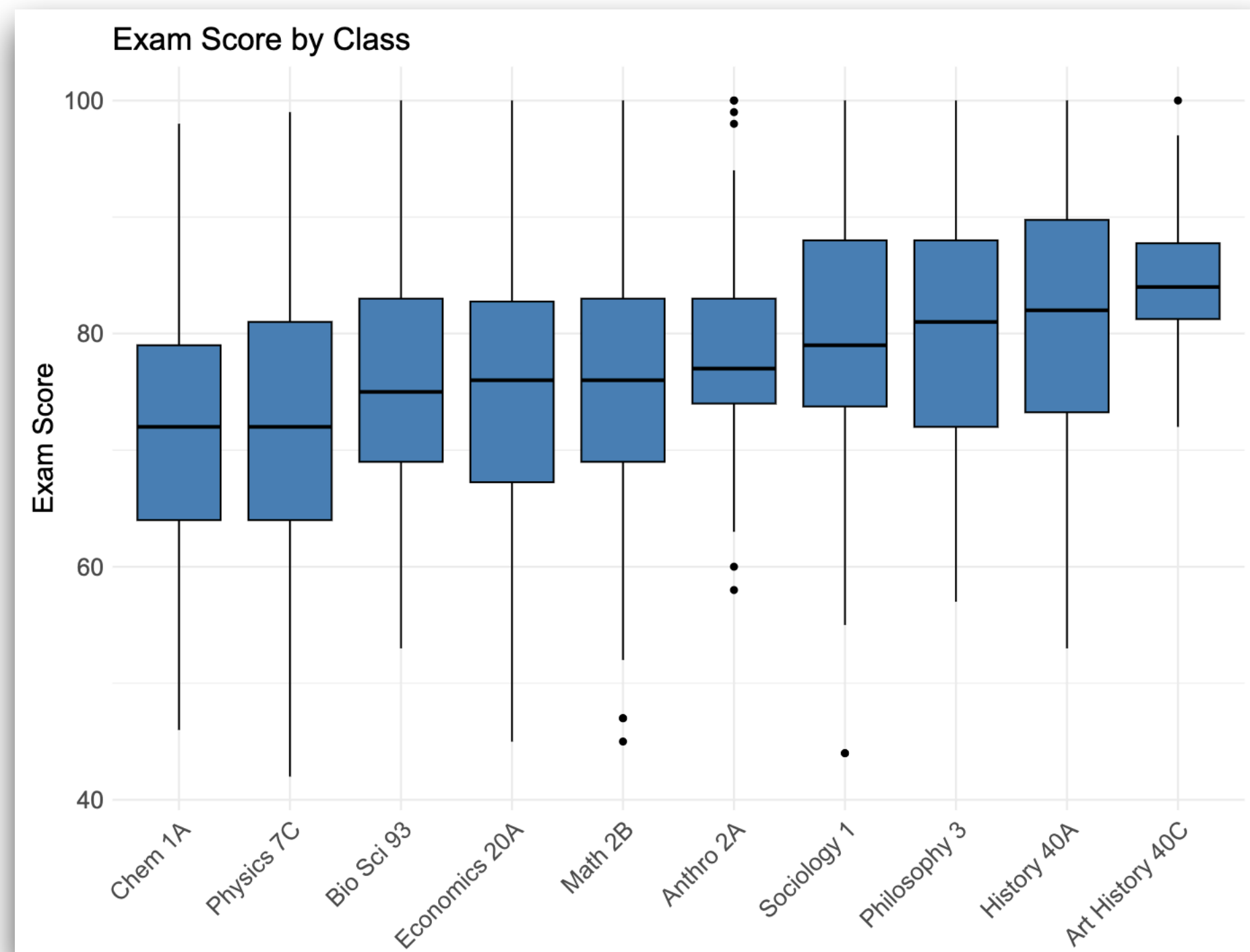


Association

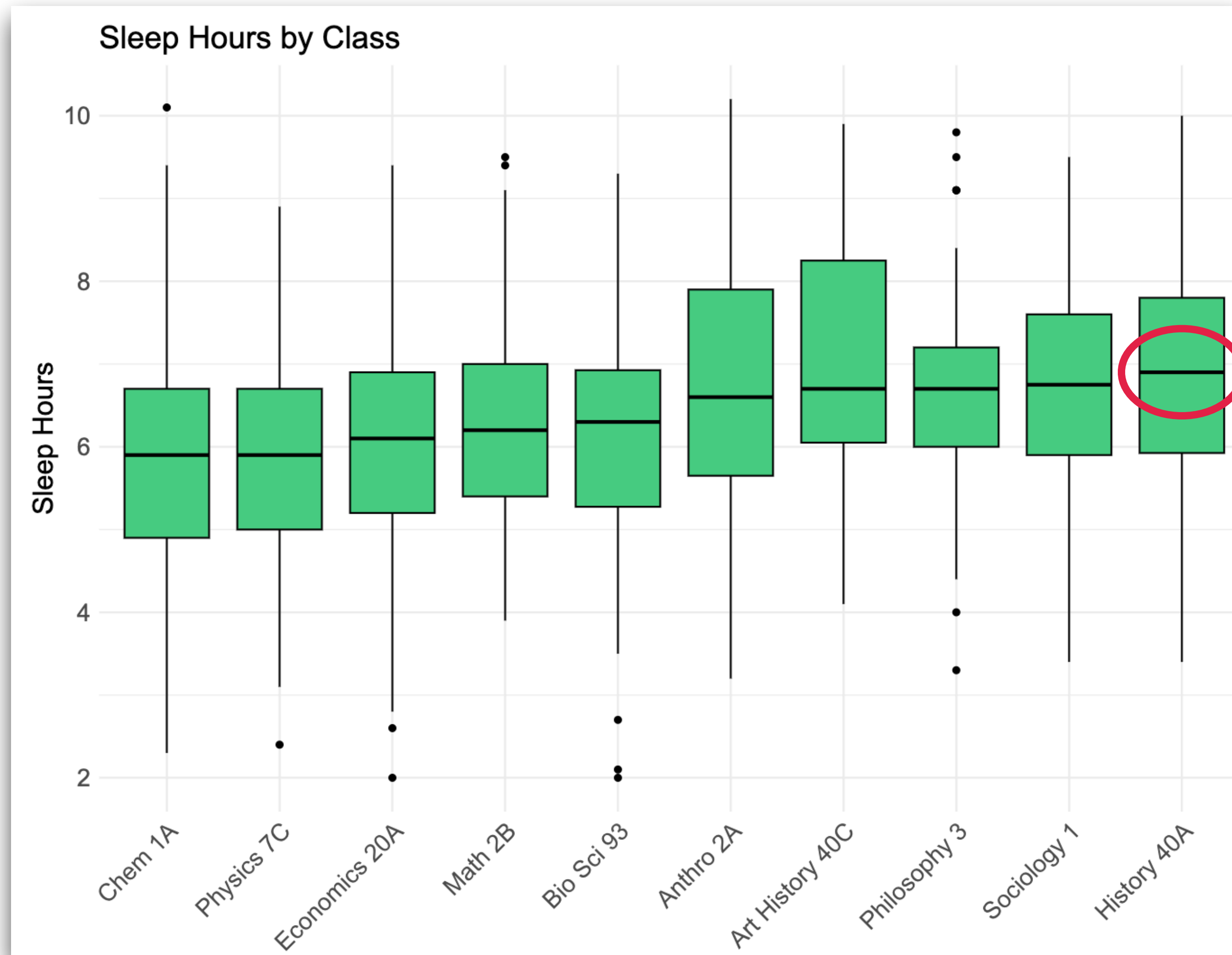
★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- We also decide to plot each quantitative variable split by class to see how different they are across classes.

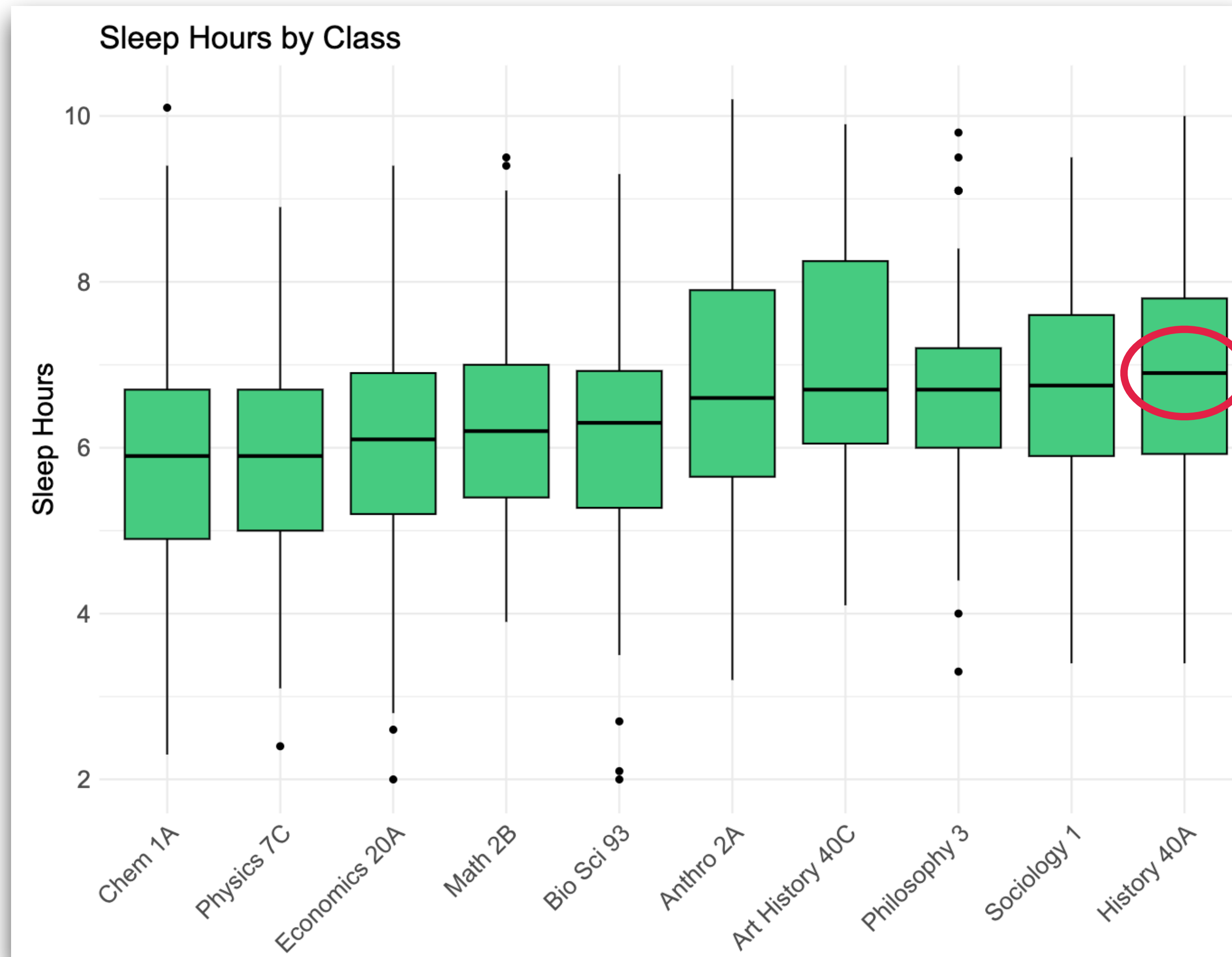


Association



20. What does this black line represent?

Association



20. What does this black line represent?

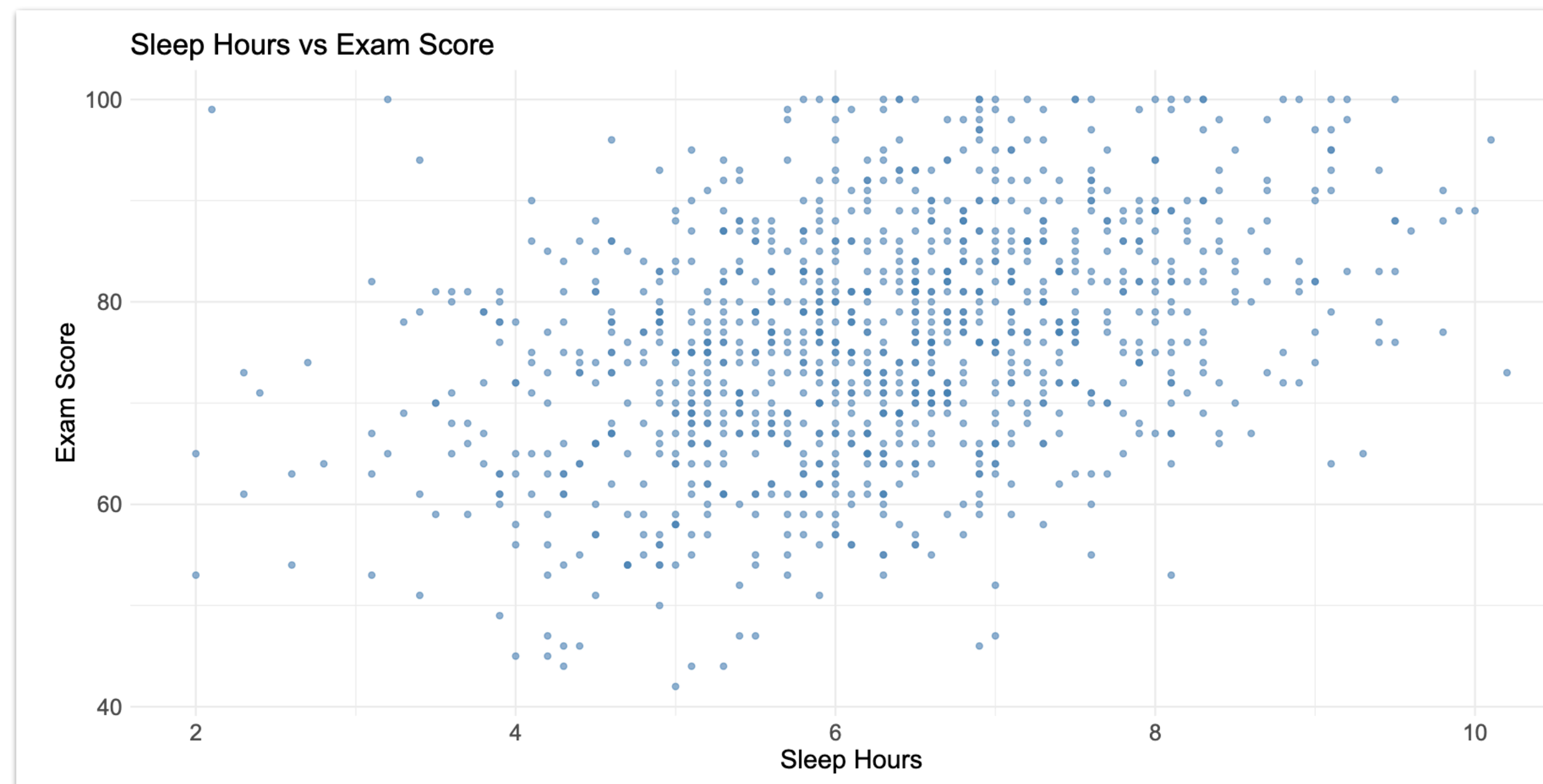
The median sleep hours the night before the exam for students in History 40A

Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- Finally, we move on to the main event: the association between sleep and exam score.



21. Is the correlation coefficient...

Positive?

Negative?

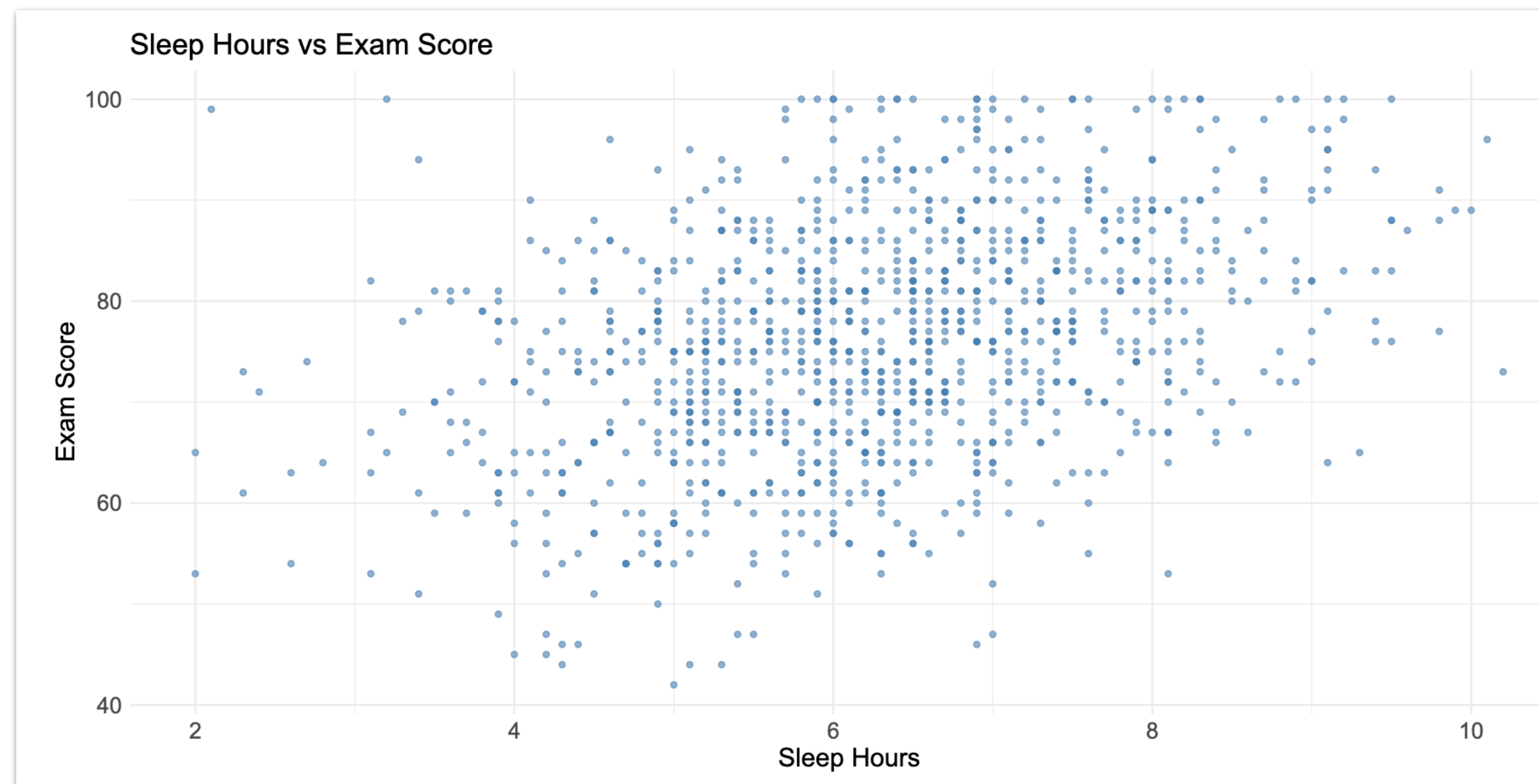
Approximately 0?

Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

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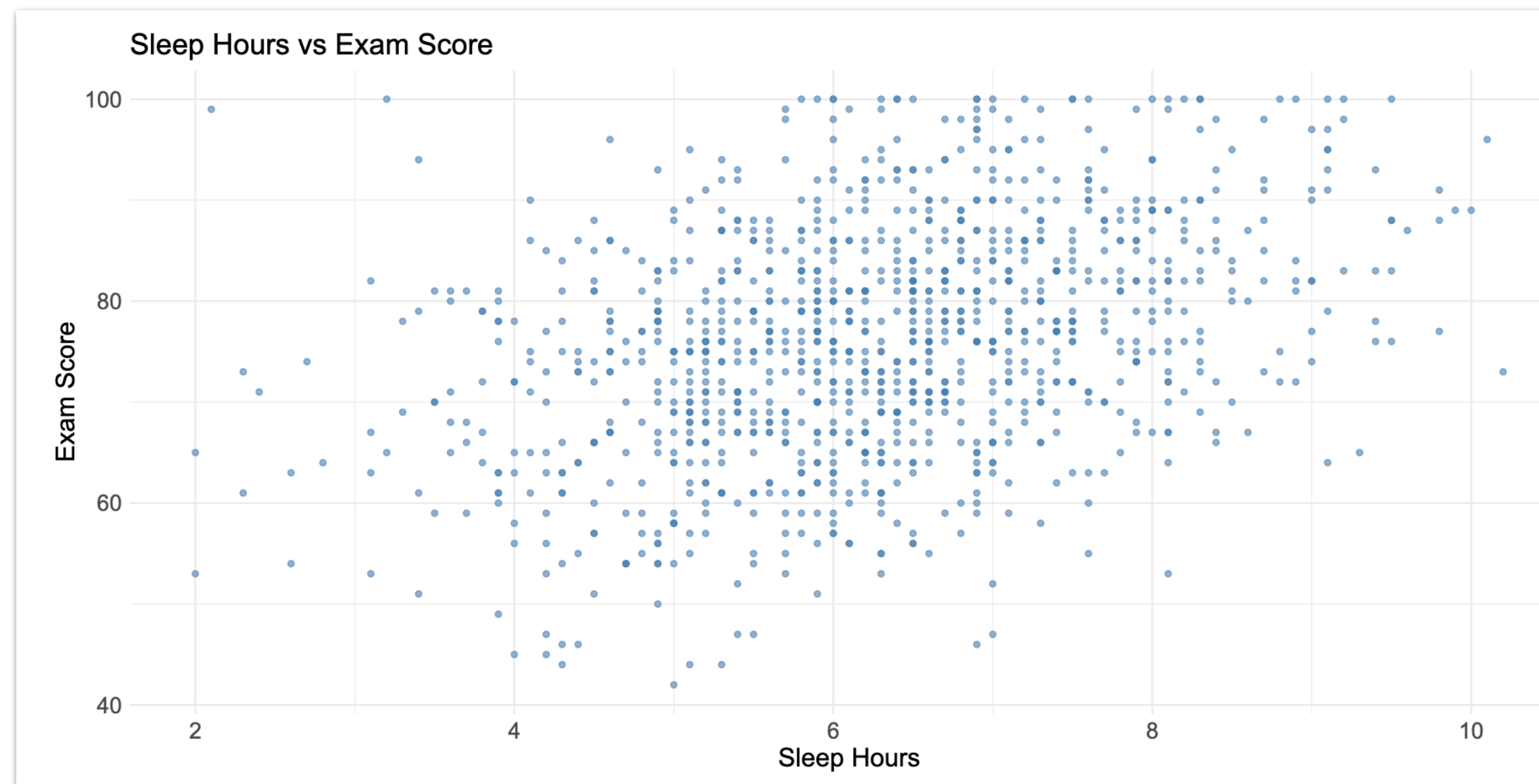
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Association

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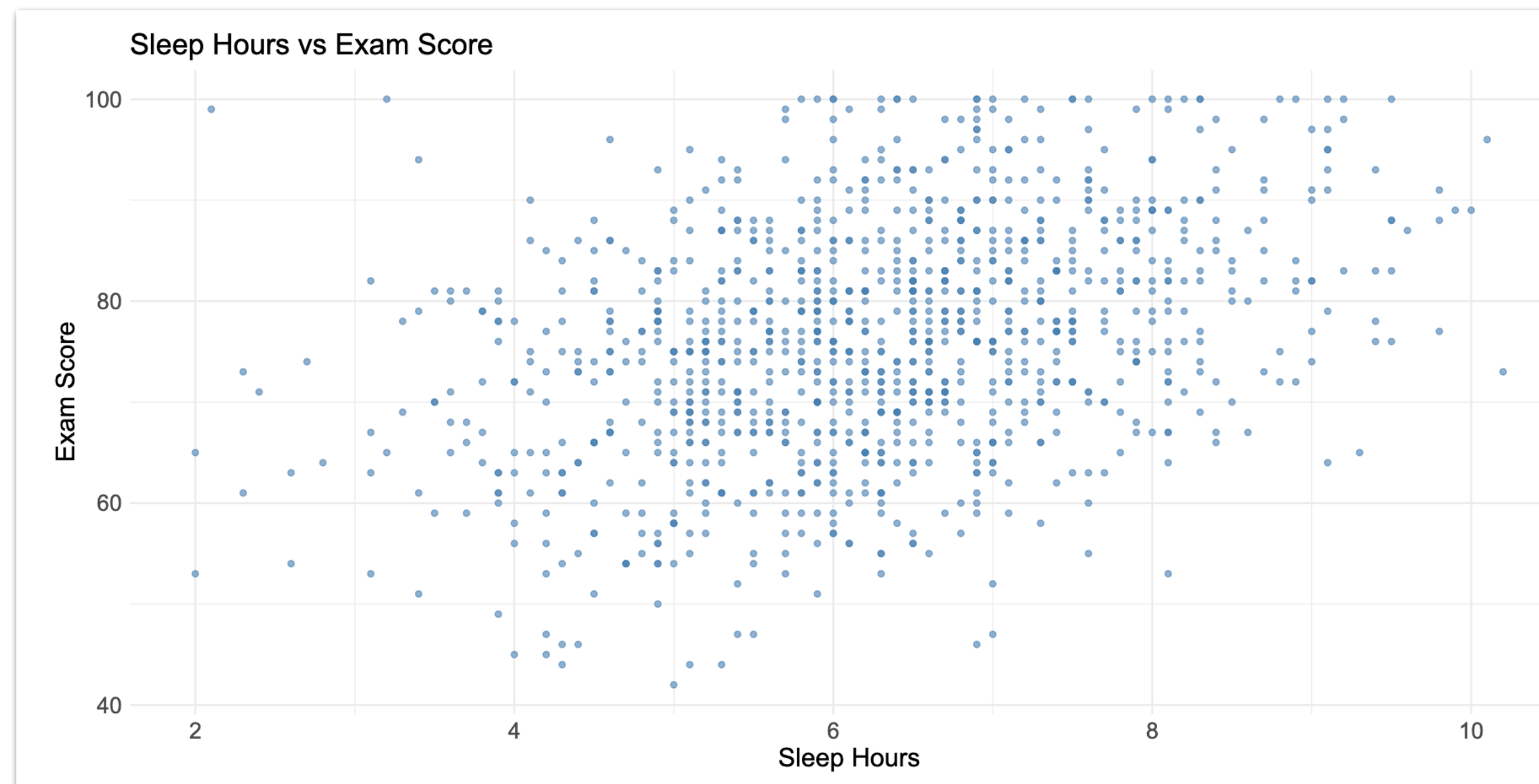
22. The correlation coefficient is **0.378**. What percentage of the variation in exam score can be explained by sleep hours?

Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- Finally, we move on to the main event: the association between sleep and exam score.



22. The correlation coefficient is **0.378**. What percentage of the variation in exam score can be explained by sleep hours?

About 14.3%

Association

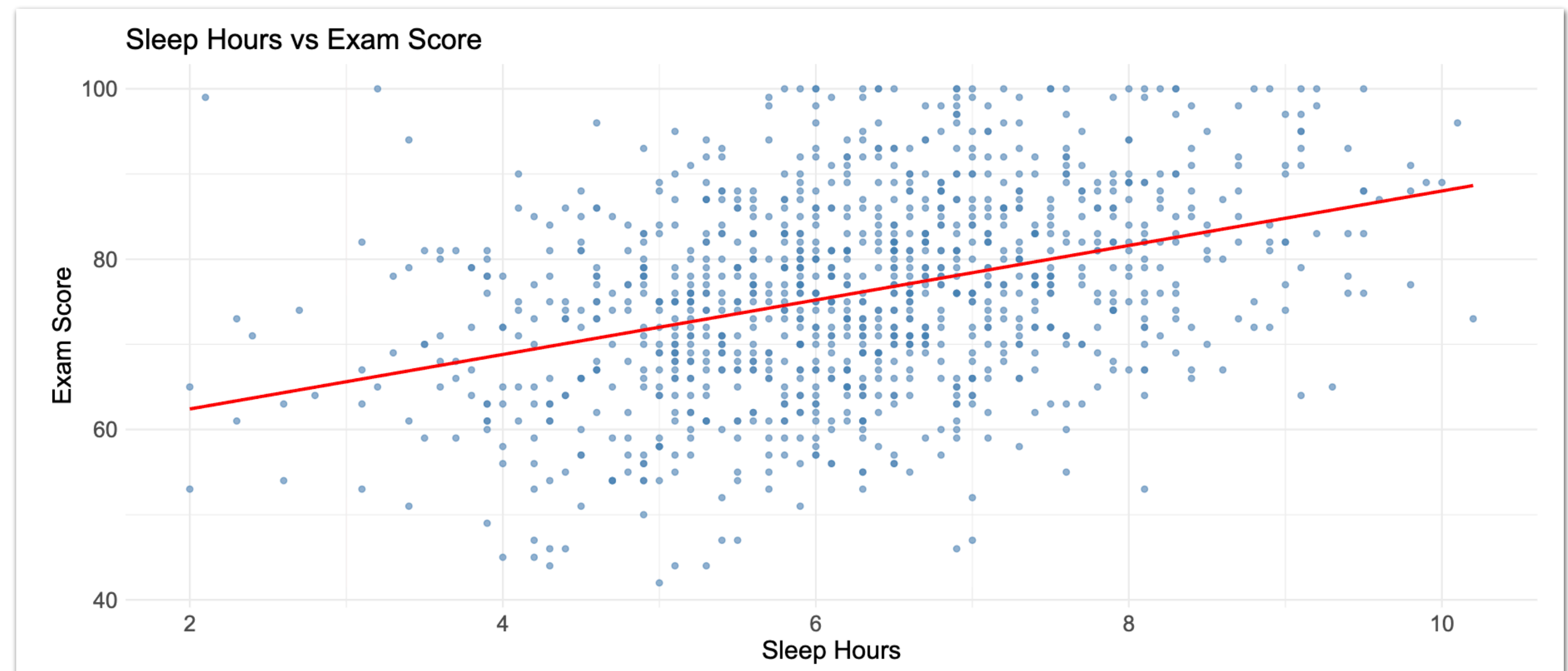
★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- The least squares regression line describing these data is given by:

$$(\text{exam score}) = 56.01 + 3.20 \times (\text{sleep hours})$$

23. How can the intercept be interpreted?



Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

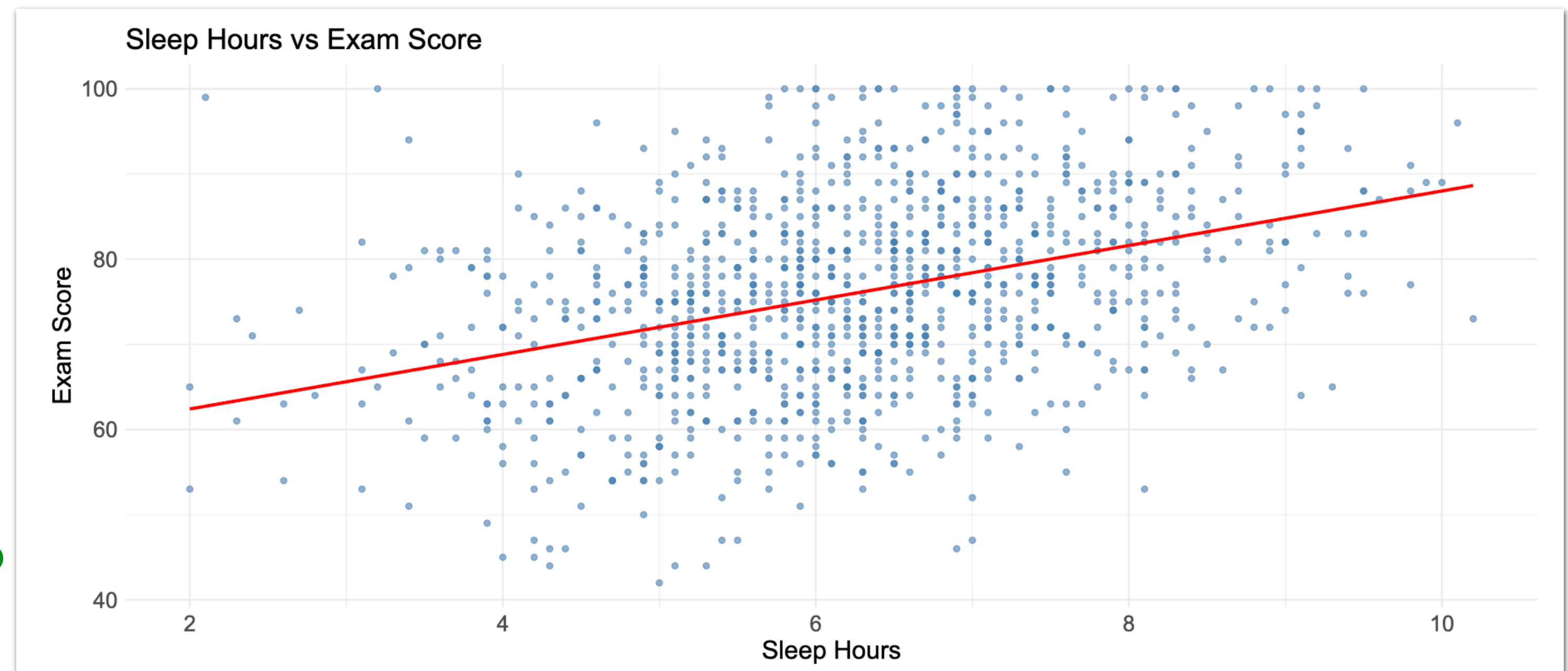
- The least squares regression line describing these data is given by:

$$(\text{exam score}) = 56.01 + 3.20 \times (\text{sleep hours})$$

23. How can the intercept be interpreted?

According to the regression line, students with 0 hours of sleep would have an average exam score of about 56.01.

This is outside the observed range, so it is not practically meaningful.



Association

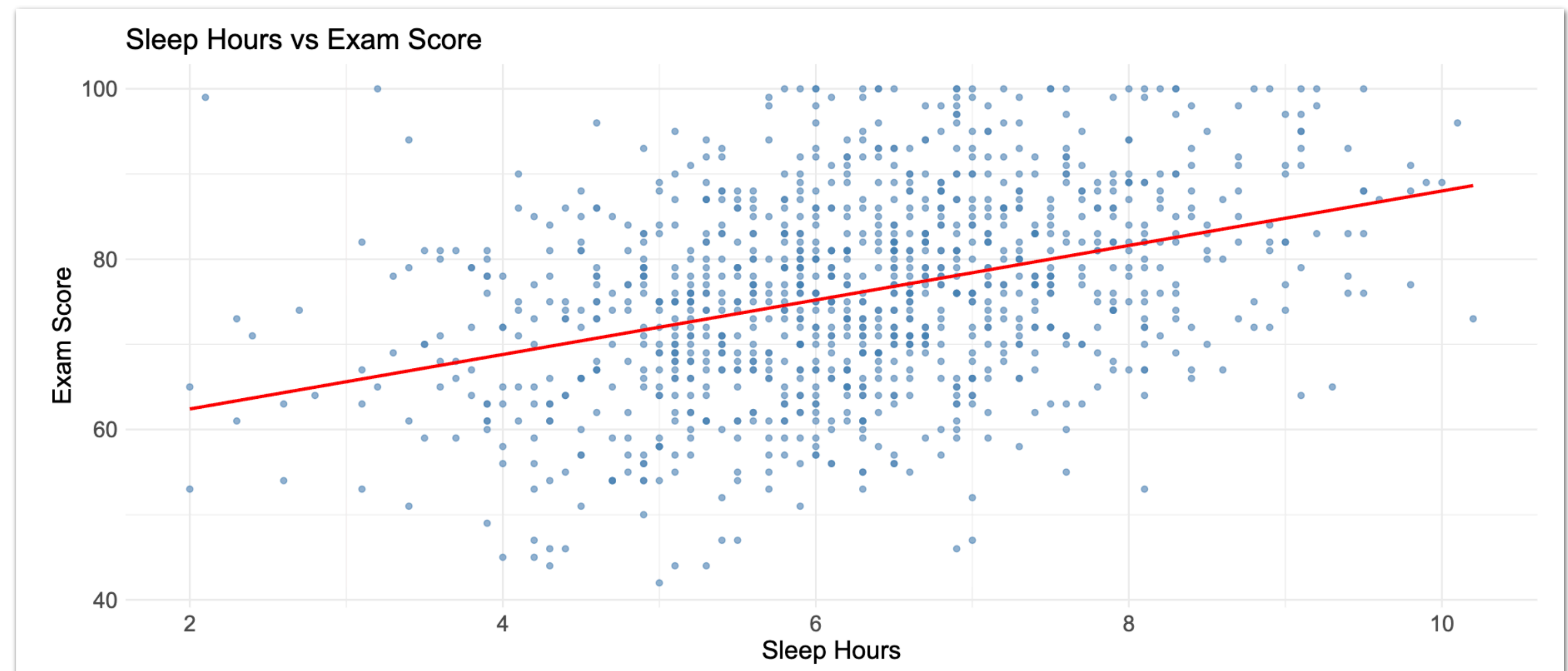
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24. How can the slope be interpreted?



Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

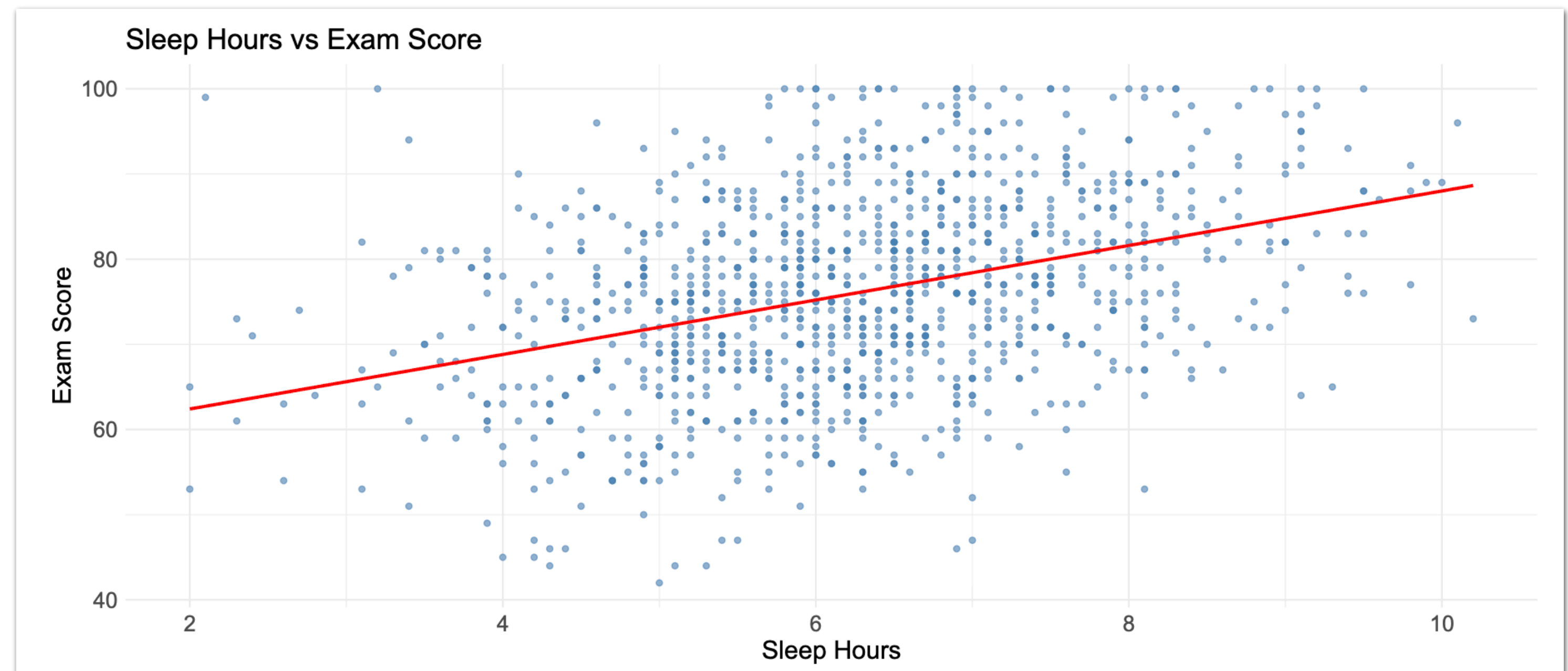
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24. How can the slope
be interpreted?

Comparing two groups of students in this study who differed by 1 hour of sleep, the group with more sleep had average exam scores about 3.20 points higher.



Association

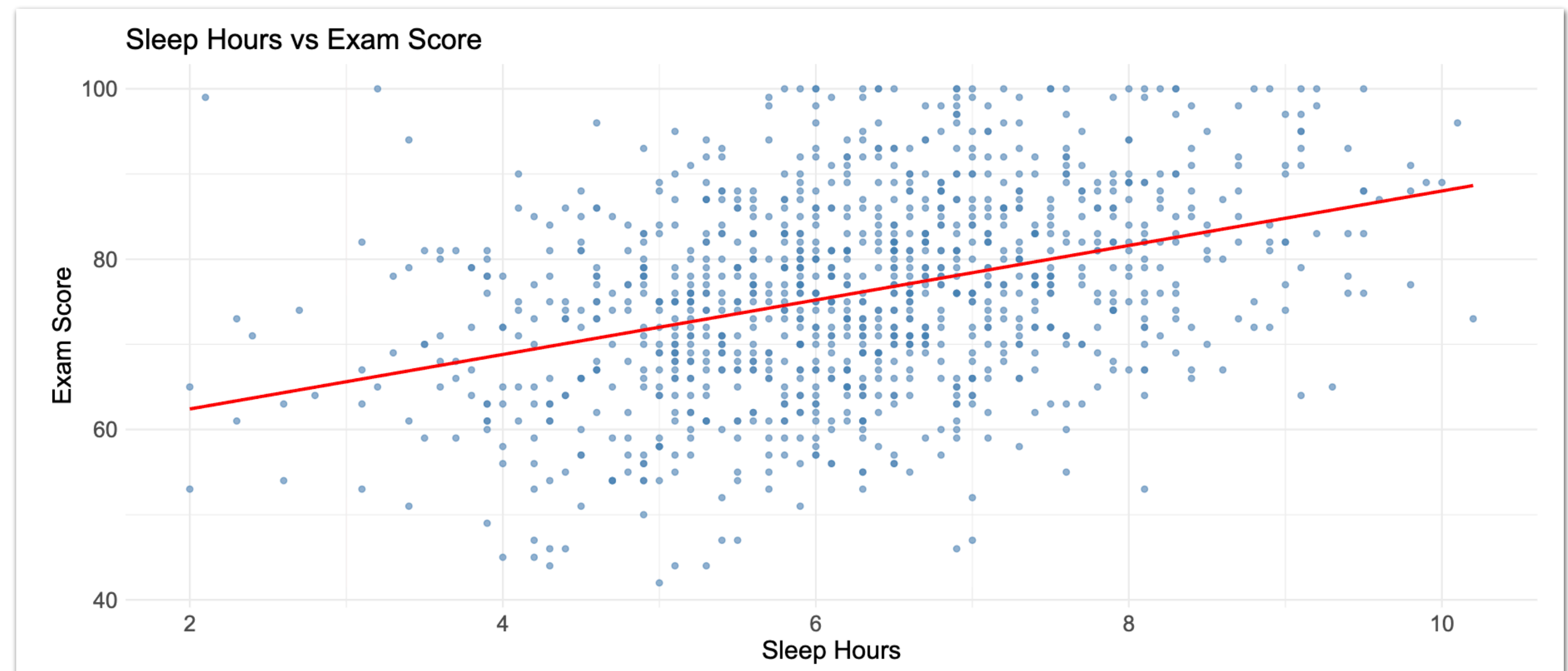
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$$(\text{exam score}) = 56.01 + 3.20 \times (\text{sleep hours})$$

25. What is the predicted exam score for a student who slept 5 hours?



Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

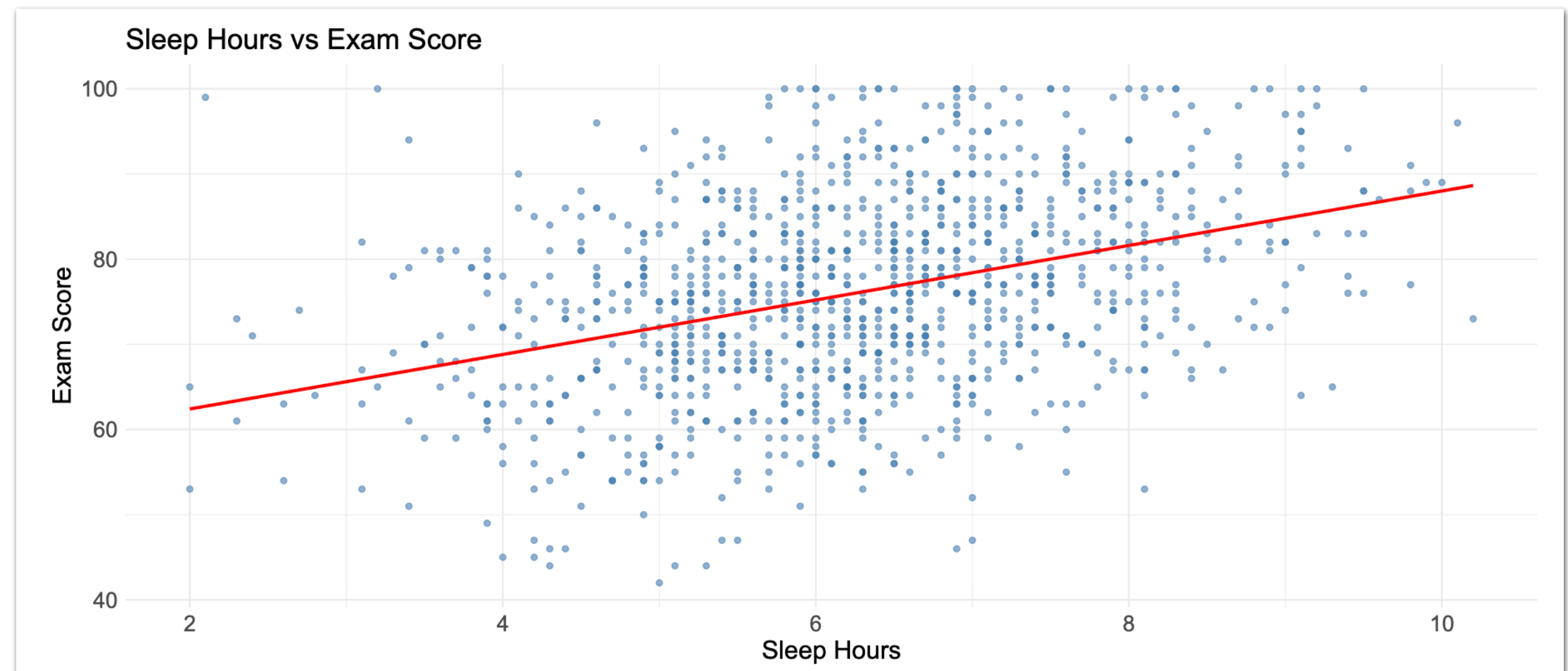
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25. What is the predicted exam score for a student who slept 5 hours?

72.01



Association

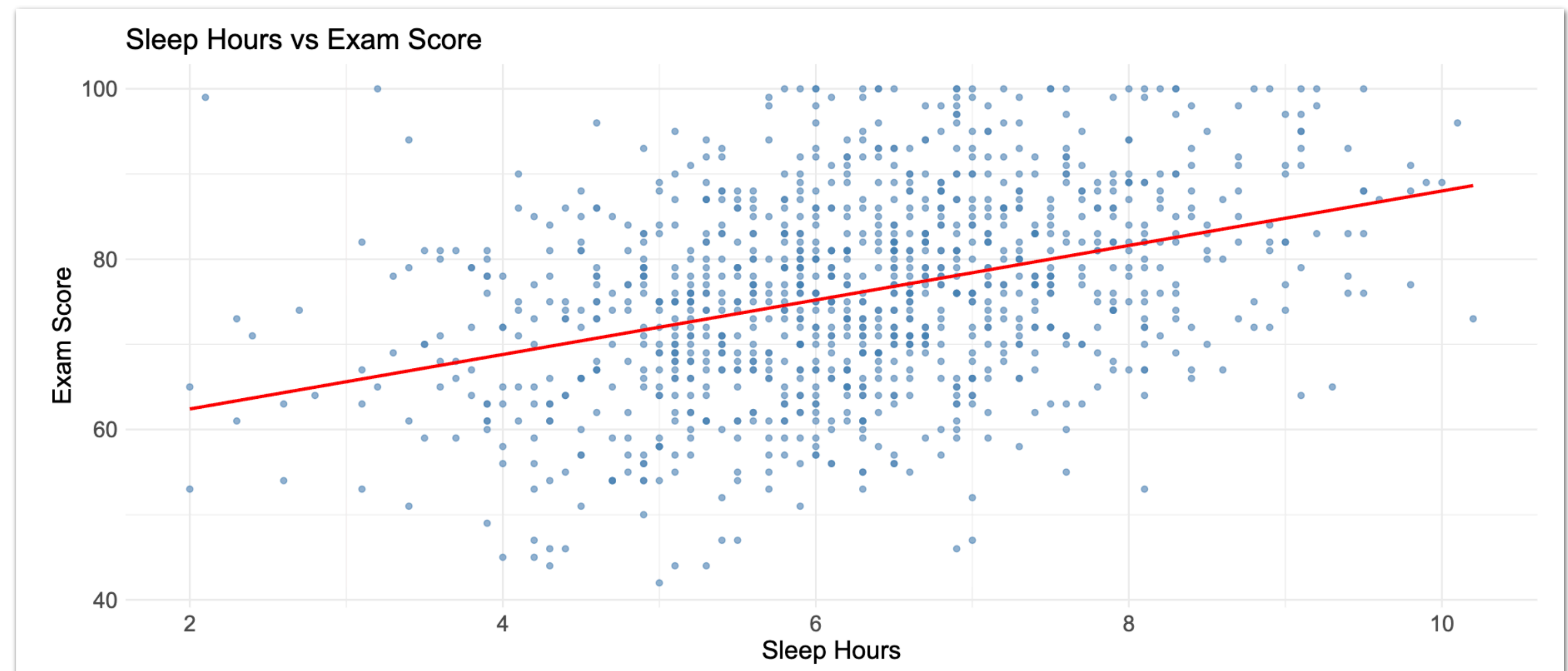
★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 Data collection: cluster sampling of undergraduate courses, anonymous survey

- The least squares regression line describing these data is given by:

$$(\text{exam score}) = 56.01 + 3.20 \times (\text{sleep hours})$$

26. What is the predicted exam score for a student who slept 1 hour?



Association

★ Is getting more sleep associated with higher exam scores in undergraduate students?

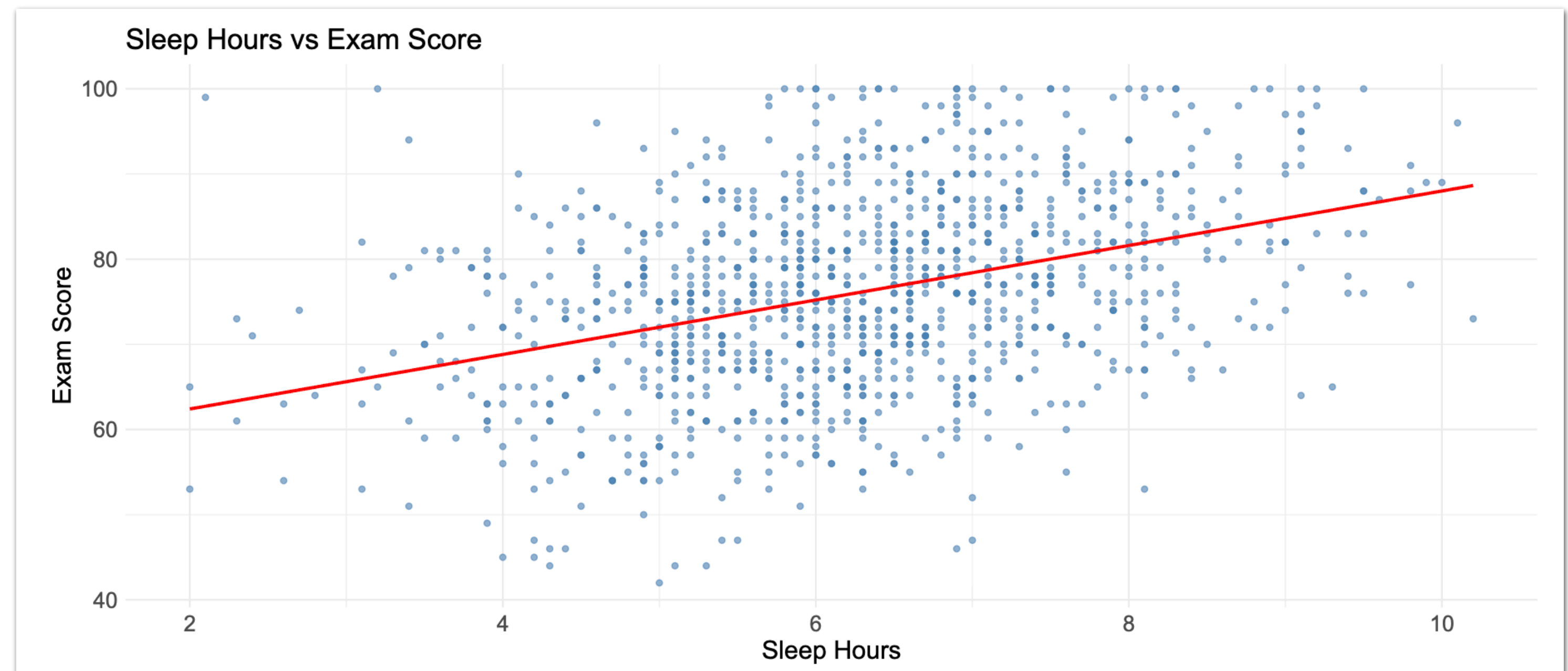
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- The least squares regression line describing these data is given by:

$$(\text{exam score}) = 56.01 + 3.20 \times (\text{sleep hours})$$

26. What is the predicted exam score for a student who slept 1 hour?

NA



Conclusions

★ Is getting more sleep associated with higher exam scores in undergraduate students?

📌 In this sample of UCI students from selected classes, there was a **positive association** between reported sleep the night before a midterm and exam score.

📌 Comparing two groups of students in this study differing in reported sleep hours by 1, the group that slept more had average reported exam scores 3.20 points higher.

📌 However, because the study was observational (so there are possibly unmeasured confounders), this does not show that more sleep causes higher exam scores.

✳️ Go to sleep early this Sunday, folks! The association *could* be causal...